

جامعة أم القرى
UMM AL-QURA UNIVERSITY

Lower Limb muscles

Dr.Yasser Al HArbi

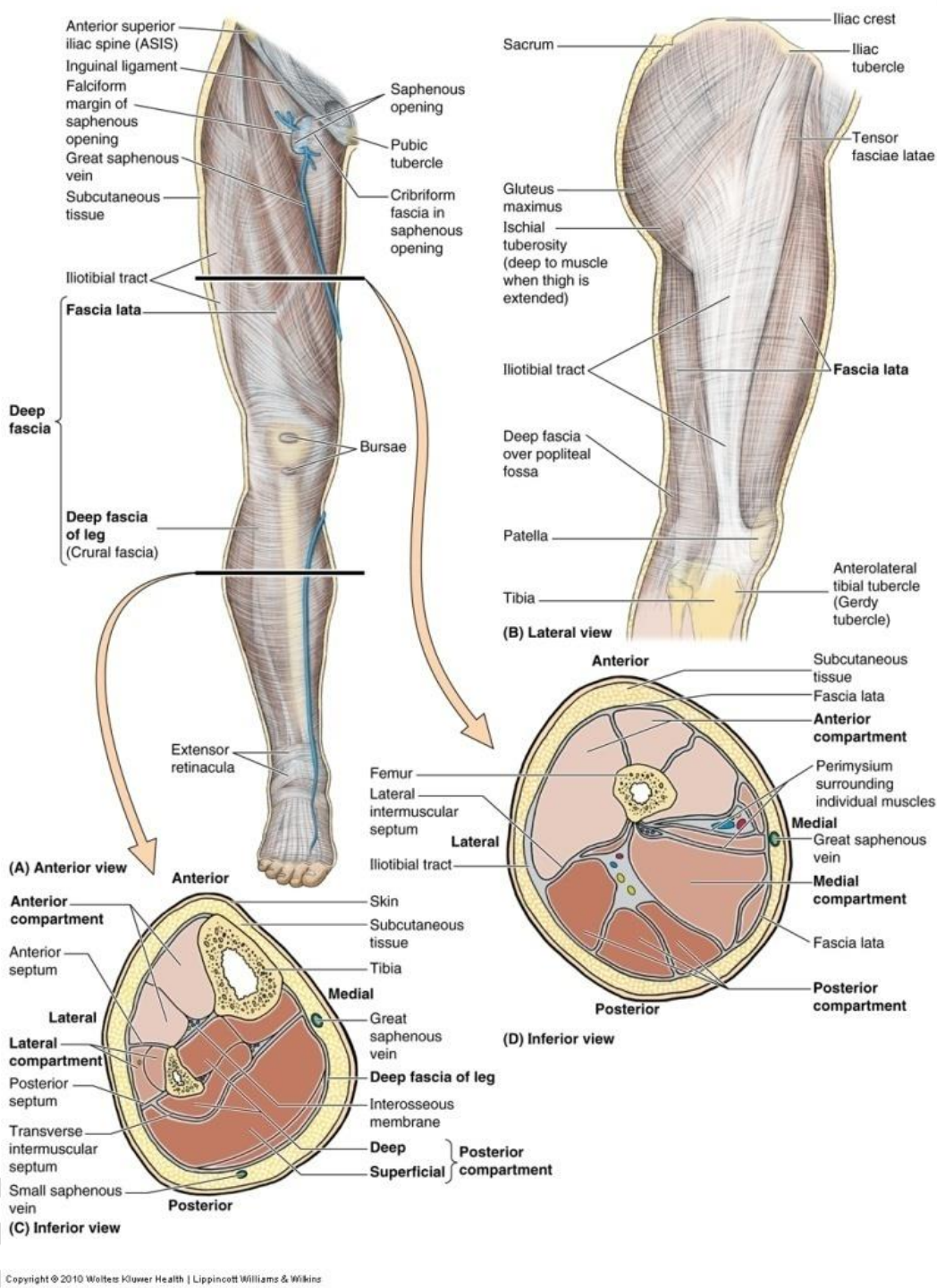
Learning objectives

By the end of the lecture the student should be able to describe the general attachments, actions and innervations of the following muscles:

- Hip-thigh: gluteus maximus, medius and minimus, tensor fascia lata , lateral rotators (e.g. piriformis, pronator quadratus, gemelli), iliopsoas, adductors (e.g. adductor magnus, longu, brevis, and gracilis), sartorius, biceps femoris, semimembranosus, semitendinosus
- Leg-foot; gastronomies, tibialis anterior, fibularis (peroneus) longus and brevis, soleus, extensor hallucis longus, plantaris, flexor digitorum longus, extensor digitorum longus, flexor hallucis longus, tibialis posterior; internal muscles of foot (as a group)

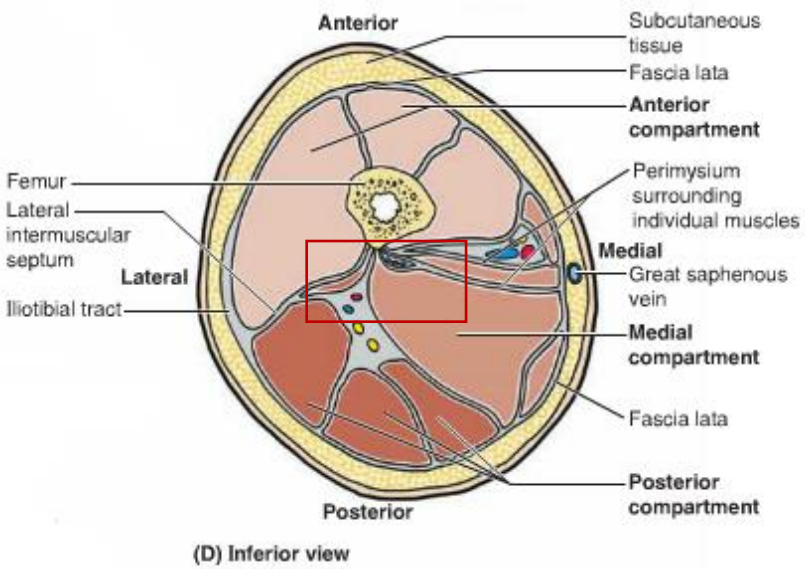
Lower limb fascia and compartments

- Fascia lata (in thigh)
- Iliotibial tract
- Saphenous opening
- Crural fascia (in leg)

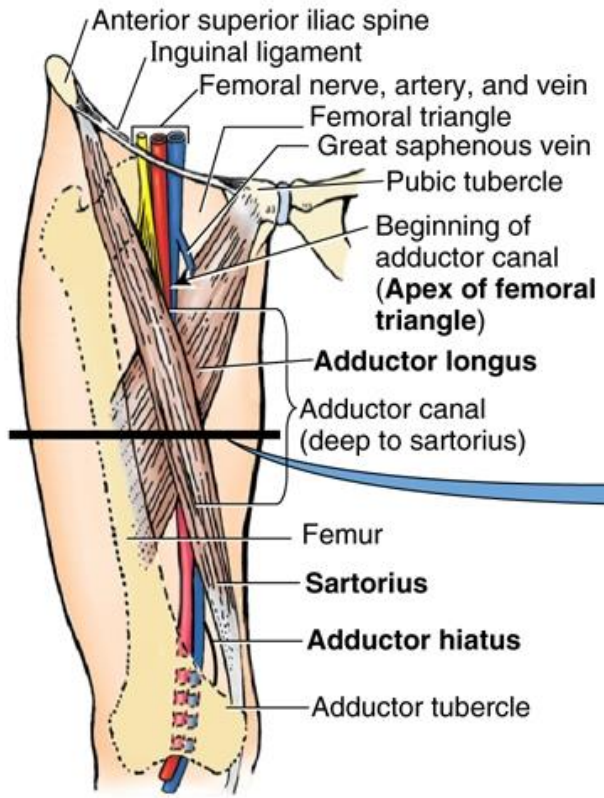


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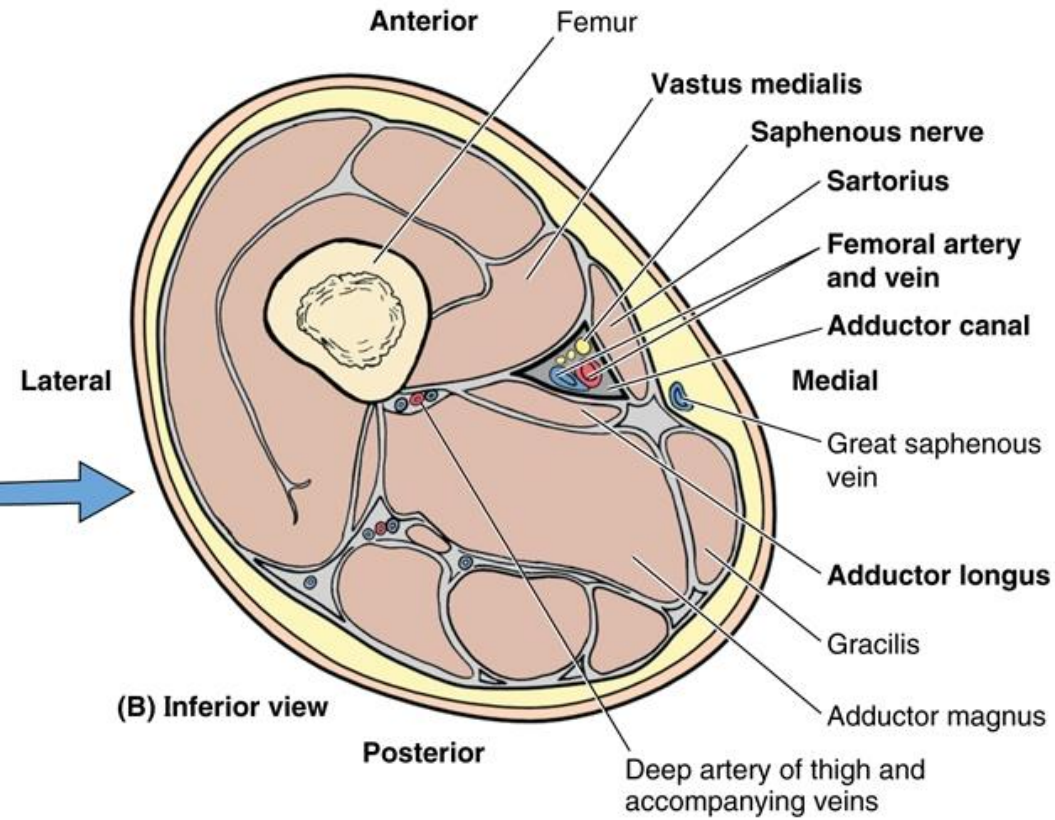
Thigh compartments



(D) Inferior view



(A) Anterior view

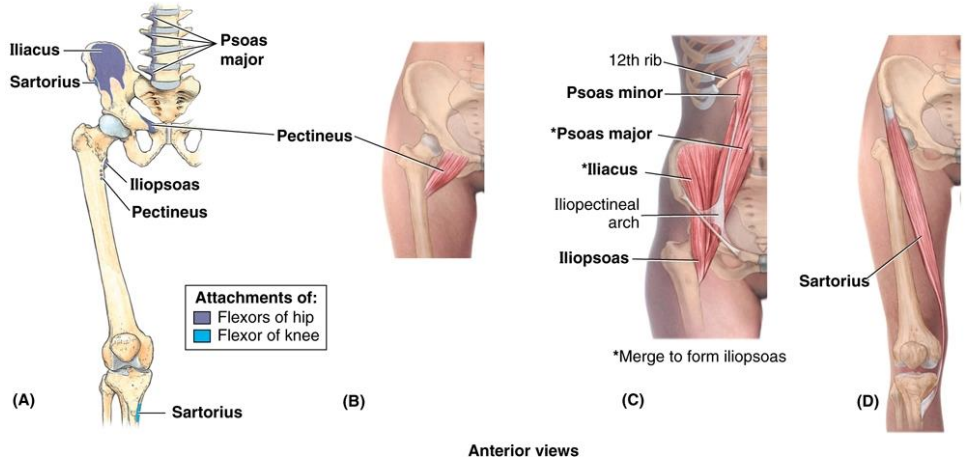


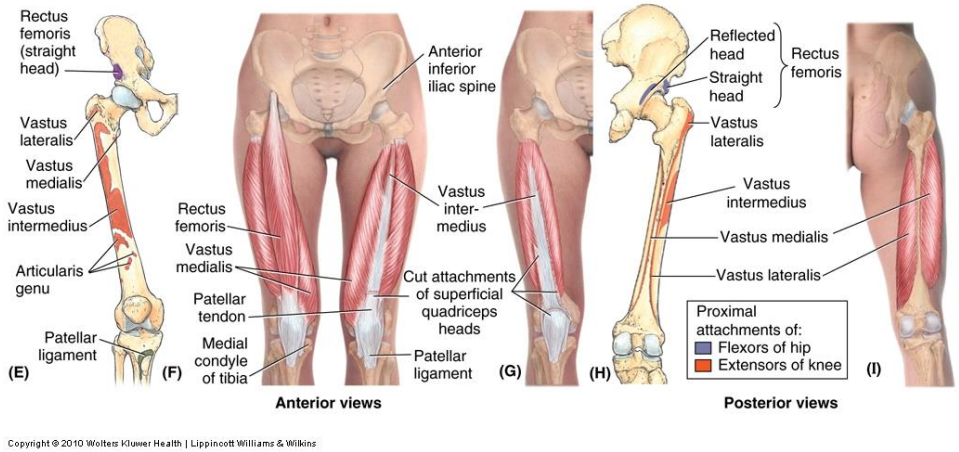
(B) Inferior view

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Anterior Thigh Muscles

- Mainly femoral nerve •
- Flexors of the hip •
  - Pectineus m. •
  - Iliopsoas m. •
  - Iliacus m. •
  - Psoas major m. •
  - Major flexor ➢
- Sartorius m. •
  - Longest •
  - Flex hip & knee •
- Extensors of the knee •
  - Quadriceps femoris m. •
    - Rectus femoris m. •
      - Can flex the hip •
    - Vastus medialis m. •
    - Vastus intermedius m. •
    - Articularis genu •
    - Vastus lateralis m. •





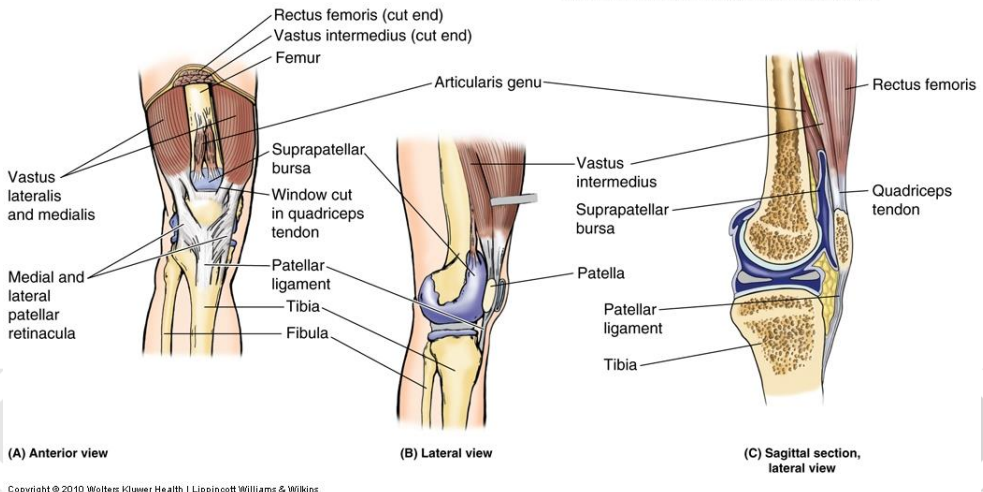


TABLE 5.3.I. MUSCLES OF ANTERIOR THIGH: FLEXORS OF HIP JOINT

Muscle	Proximal Attachment[a]	Distal Attachment	Innervation[b]	Main Action(s)
Pectineus (Fig. 5.21A & B)	Superior ramus of pubis	Pectineal line of femur, just inferior to lesser trochanter	Femoral nerve (**L2,** L3); may receive a branch from obturator nerve	Adducts and flexes thigh; assists with medial rotation of thigh
Iliopsoas (Fig. 5.21A & C) Psoas major	Sides of T12–L5 vertebrae and discs between them; transverse processes of all lumbar vertebrae	Lesser trochanter of femur	Anterior rami of lumbar nerves (**L1, L2,** L3)	Act conjointly in flexing thigh at hip joint and in stabilizing this joint[c]
Psoas minor	Sides of T12–L1 vertebrae and intervertebral discs	Pectineal line, iliopectineal eminence via iliopectineal arch	Anterior rami of lumbar nerves (L1, L2)	
Iliacus	Iliac crest, iliac fossa, ala of sacrum, and anterior sacroiliac ligaments	Tendon of psoas major, lesser trochanter, and femur distal to it	Femoral nerve (**L2,** L3)	
Sartorius (Fig. 5.21A & D)	Anterior superior iliac spine and superior part of notch inferior to it	Superior part of medial surface of tibia	Femoral nerve (L2, L3)	Flexes, abducts, and laterally rotates thigh at hip joint; flexes leg at knee joint, (medially rotating leg when knee is flexed)[d]

[a] The Latin word *insertio* means *attachment*. The terms insertion and origin (L. *origo*) have not been used here (or elsewhere) since they change with function.

[b] The spinal cord segmental innervation is indicated (e.g., "L1, L2, L3" means that the nerves supplying the psoas major are derived from the first three lumbar segments of the spinal cord). Numbers in boldface (**L1, L2**) indicate the main segmental innervation. Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

[c] The psoas major is also a postural muscle that helps control the deviation of the trunk and is active during standing.

[d] The four actions of the sartorius (L. *sartor,* tailor) produce the once common cross-legged sitting position used by tailors, hence the name.

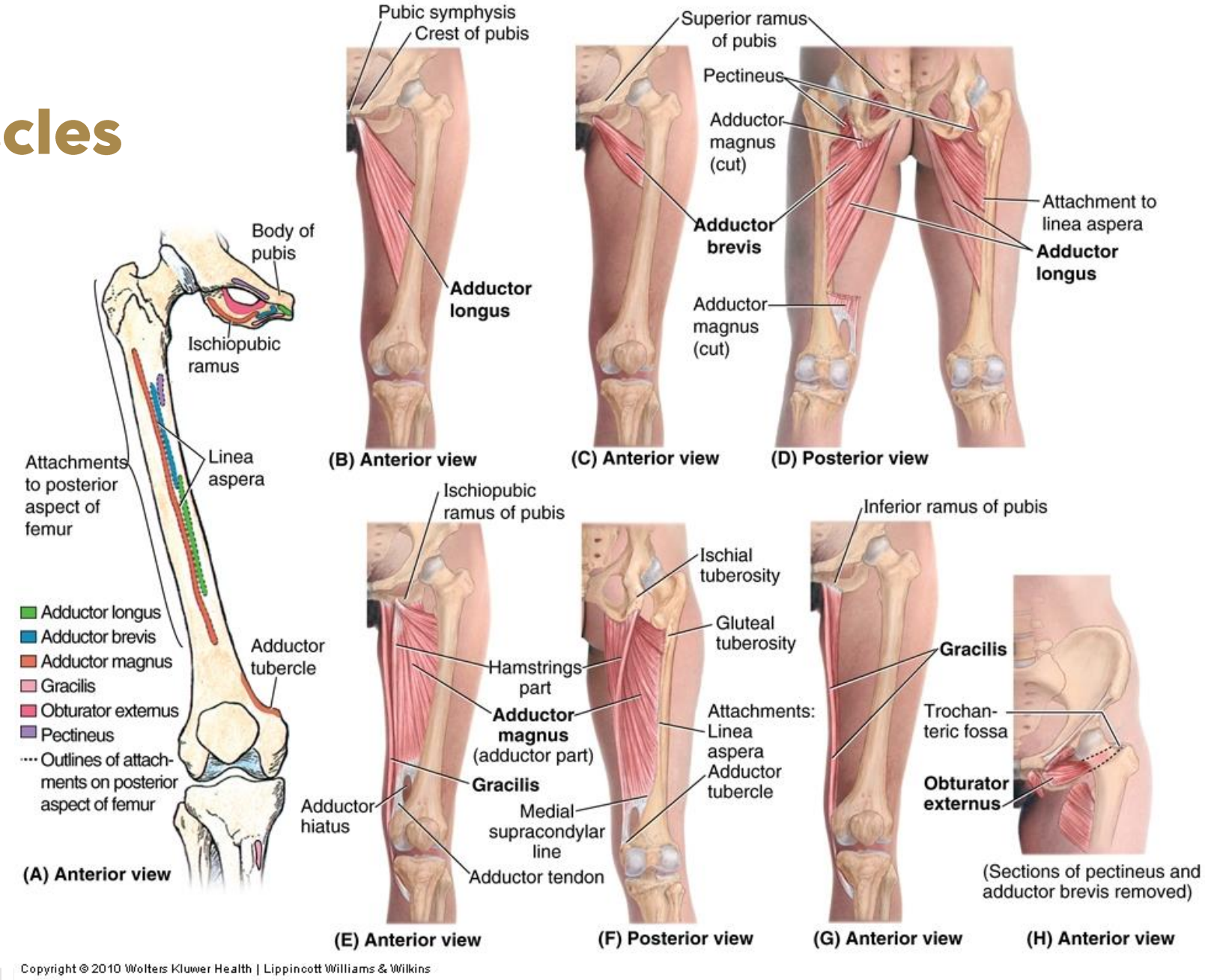
TABLE 5.3.II. MUSCLES OF ANTERIOR THIGH: EXTENSORS OF KNEE

Muscle	Proximal Attachment	Distal Attachment	Innervation[a]	Main Action
Quadriceps femoris (Fig. 5.21E–H)				
Rectus femoris	Anterior inferior iliac spine and ilium superior to acetabulum	Via common tendinous (quadriceps tendon) and independent attachments to base of patella; indirectly via patellar ligament to tibial tuberosity; medial and lateral vasti also attach to tibia and patella via aponeuroses (medial and lateral patellar retinacula)	Femoral nerve (L2, **L3, L4**)	Extend leg at knee joint; rectus femoris also steadies hip joint and helps iliopsoas flex thigh
Vastus lateralis	Greater trochanter and lateral lip of linea aspera of femur			
Vastus medialis	Intertrochanteric line and medial lip of linea aspera of femur			
Vastus intermedius	Anterior and lateral surfaces of shaft of femur			

[a] The spinal cord segmental innervation is indicated (e.g., "L1, L2, L3" means that the nerves supplying the quadriceps femoris are derived from the first three lumbar segments of the spinal cord). Numbers in boldface (**L3, L4**) indicate the main segmental innervation. Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

Medial Thigh Muscles

- Mainly obturator nerve ➢
- Adductor group ➢
  - Adductor longus m. •
    - Most anterior •
  - Adductor brevis m. •
    - Deep to pectinus & adductor longus •
  - Adductor magnus m. •
    - Most posterior •
  - Adductor hiatus •
  - Adductor part ➢
  - Hamstring part ➢
    - Sciatic nerve ➢
  - Gracilis m. •
    - Most medial and superficial •
    - Crosses hip & knee joints •
  - Obturator externus m. •



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TABLE 5.4. MUSCLES OF MEDIAL THIGH: ADDUCTORS OF THIGH

Muscle[a]	Proximal Attachment	Distal Attachment	Innervation[b]	Main Action
Adductor longus (Fig. 5.23E & G)	Body of pubis inferior to pubic crest	Middle third of linea aspera of femur	Obturator nerve, branch of, anterior division (L2, **L3,** L4)	Adducts thigh
Adductor brevis (Fig. 5.23F & G)	Body and inferior ramus of pubis	Pectineal line and proximal part of linea aspera of femur		Adducts thigh; to some extent flexes it
Adductor magnus (Fig. 5.23C, D, & G)	Adductor part: inferior ramus of pubis, ramus of ischium Hamstrings part: ischial tuberosity	Adductor part: gluteal tuberosity, linea aspera, medial supracondylar line Hamstring part: adductor tubercle of femur	Adductor part: obturator nerve (L2, **L3, L4**), branches of posterior division Hamstring part: tibial part of sciatic nerve (**L4**)	Adducts thigh Adductor part: flexes thigh Hamstrings part: extends thigh
Gracilis (Fig. 5.23H)	Body and inferior ramus of pubis	Superior part of medial surface of tibia.	Obturator nerve (**L2,** L3)	Adducts thigh; flexes leg; helps rotate leg medially
Obturator externus	Margins of obturator foramen and obturator membrane	Trochanteric fossa of femur	Obturator nerve (L3, **L4**)	Laterally rotates thigh; steadies head of femur in acetabulum

[a] Collectively, the five muscles listed are the adductors of the thigh, but their actions are more complex (e.g., they act as flexors of the hip joint during flexion of the knee joint and are active during walking).

[b] The spinal cord segmental innervation is indicated (e.g., "L2, L3, L4" means that the nerves supplying the adductor longus are derived from the second to fourth lumbar segments of the spinal cord). Numbers in boldface (**L3**) indicate the main segmental innervation. Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

Femoral triangle •

Boundaries •

Content •

Femoral sheath •

Content •

Femoral canal •

Femoral ring •

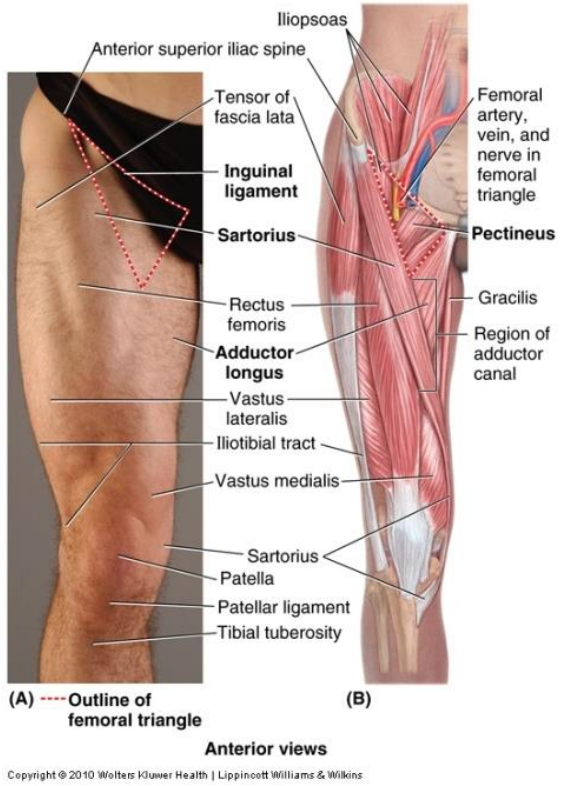
Femoral septum •

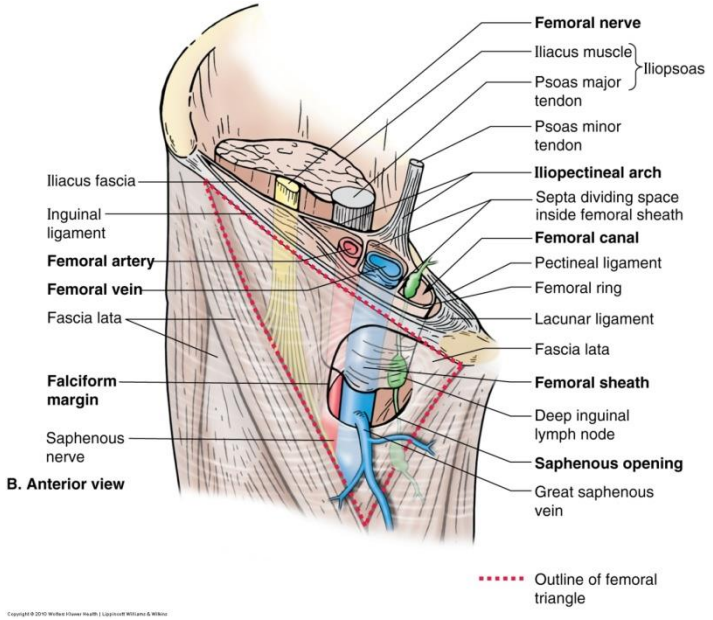
Adductor (subsartorial) canal •

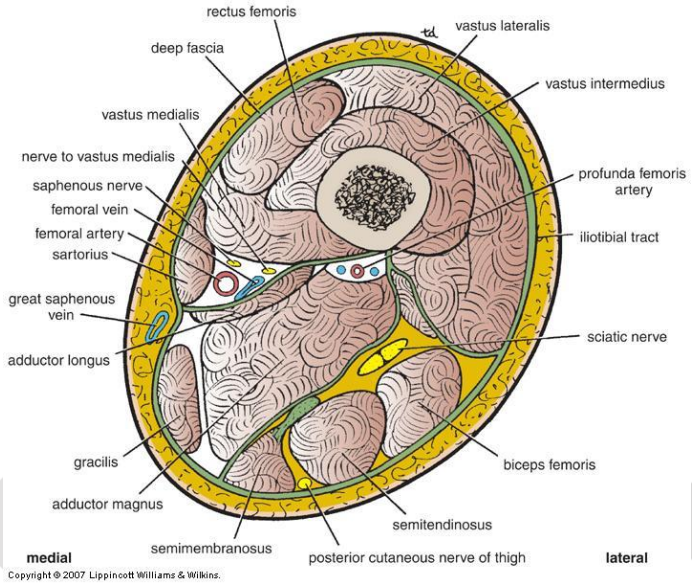
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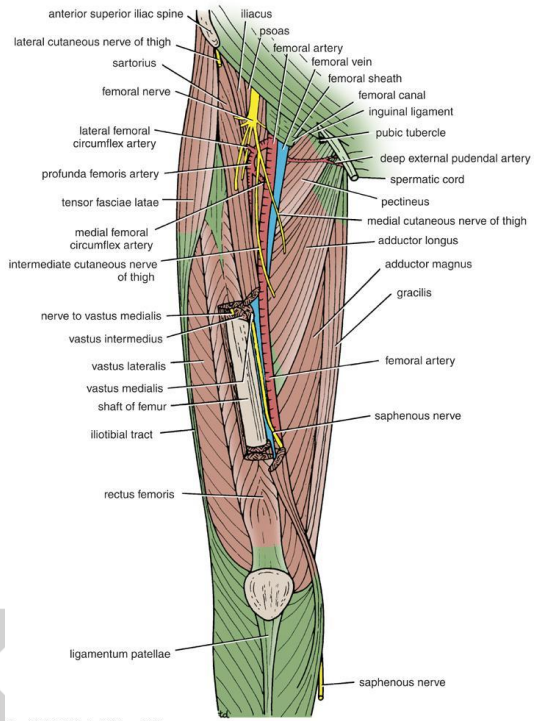
Boundaries •

content •



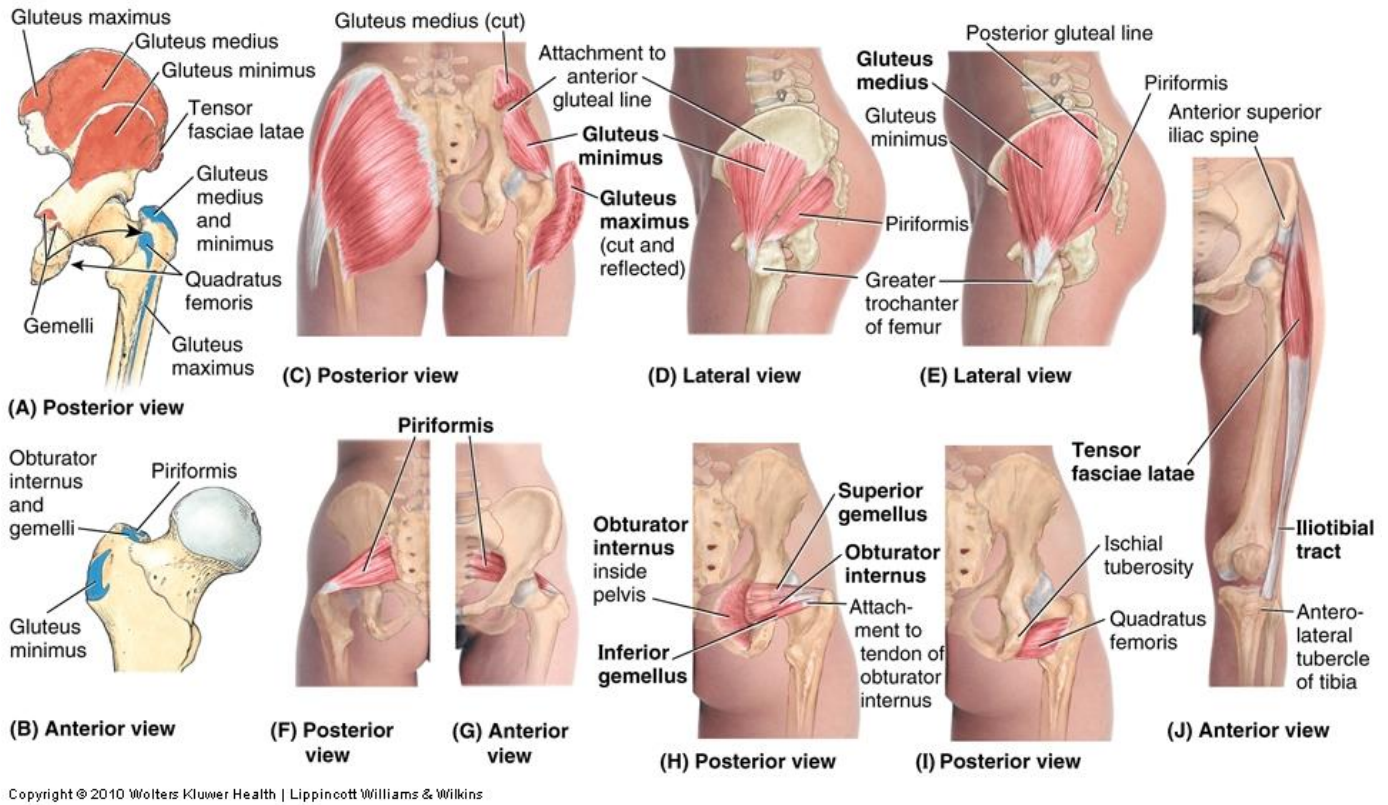


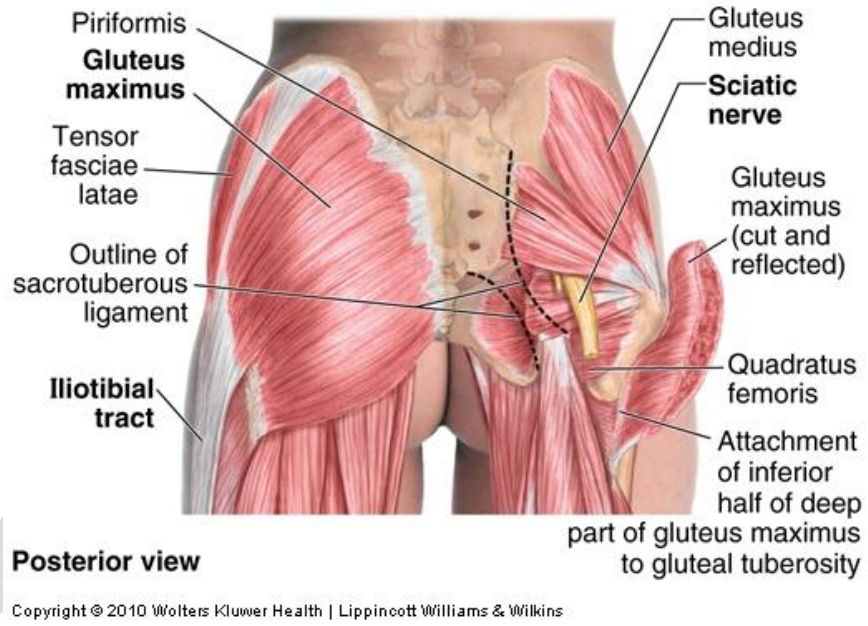




Muscles of Glutial Region

- Superficial layer
  - Extensors, abductors
- Gluteus maximus m.
  - Most superficial
  - Largest
- Gluteus medius m.
- Gluteus minimus m.
- Tenser fascialata m.





Muscles of Glutial Region

Deep layer •

Lateral rotators➢

Piriformis m. •

Greater sciatic foramen •

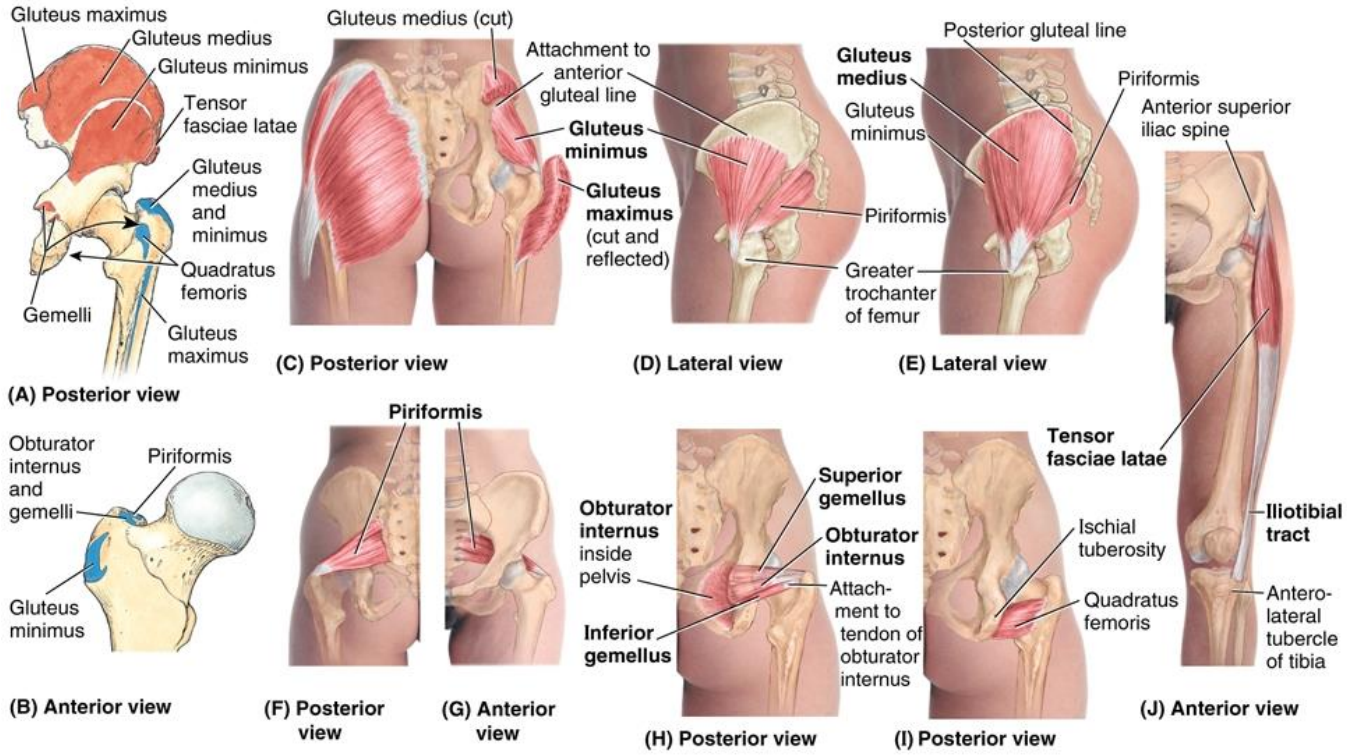
Obturator internus m. •

Lesser sciatic foramen •

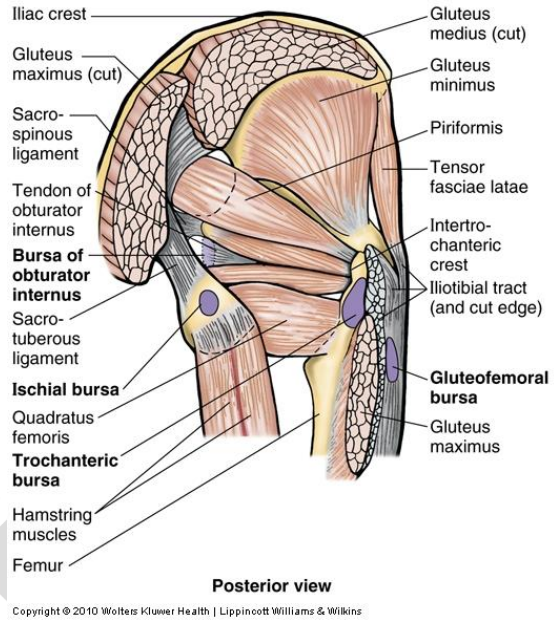
Superior gemellus m. •

Inferior gemellus m. •

Quaderatus femoris m. •



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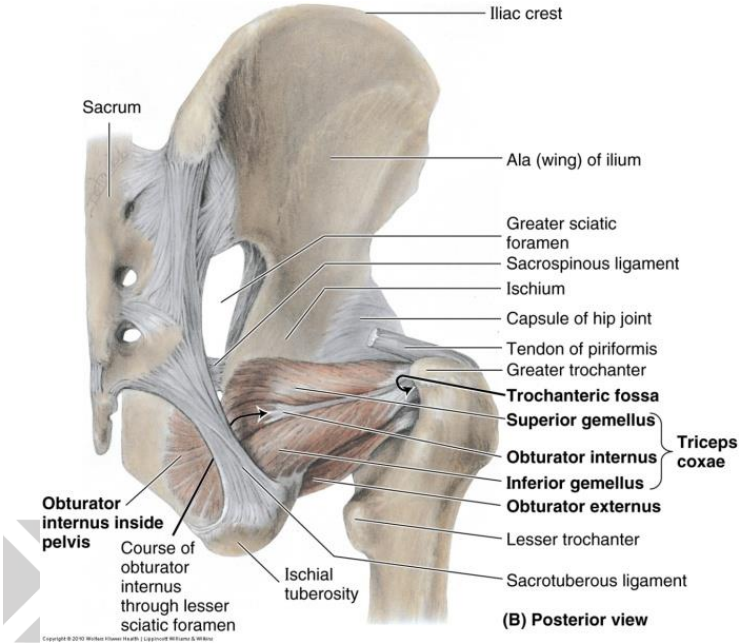


TABLE 5.6. MUSCLES OF GLUTEAL REGION: ABDUCTORS AND ROTATORS OF THIGH

Muscle	Proximal Attachment	Distal Attachment	Innervation[a]	Main Action
Gluteus maximus (Fig. 5.34A & C)	Ilium posterior to posterior gluteal line; dorsal surface of sacrum and coccyx; sacrotuberous ligament	Most fibers end in iliotibial tract, which inserts into lateral condyle of tibia; some fibers insert on gluteal tuberosity	Inferior gluteal nerve (L5, **S1, S2**)	Extends thigh (especially from flexed position) and assists in its lateral rotation; steadies thigh and assists in rising from sitting position
Gluteus medius (Fig. 5.34A, C, & E)	External surface of ilium between anterior and posterior gluteal lines	Lateral surface of greater trochanter of femur	Superior gluteal nerve (**L5,** S1)	Abduct and medially rotate thigh; keep pelvis level when ipsilateral limb is weight-bearing and advance opposite (unsupported) side during its swing phase
Gluteus minimus (Fig. 5.34A–D)	External surface of ilium between anterior and inferior gluteal lines	Anterior surface of greater trochanter of femur		
Tensor of fascia lata (Fig. 5.34J)	Anterior superior iliac spine; anterior part of iliac crest	Iliotibial tract, which attaches to lateral condyle of tibia		
Piriformis (Fig. 5.34F & G)	Anterior surface of sacrum; sacrotuberous ligament	Superior border of greater trochanter of femur	Branches of anterior rami of **S1,** S2	Laterally rotate extended thigh and abduct flexed thigh; steady femoral head in acetabulum
Obturator internus (Fig. 5.34H)	Pelvic surface of obturator membrane and surrounding bones	Medial surface of greater trochanter (trochanteric fossa) of femur[b]	Nerve to obturator internus (L5, **S1**)	
Superior and inferior gemelli (Fig. 5.34H)	Superior: ischial spine Inferior: ischial tuberosity	Medial surface of greater trochanter (trochanteric fossa) of femur[b]	Superior gemellus: same nerve supply as obturator internus Inferior gemellus: same nerve supply as quadratus femoris	
Quadratus femoris (Fig. 5.34I)	Lateral border of ischial tuberosity	Quadrate tubercle on intertrochanteric crest of femur and area inferior to it	Nerve to quadratus femoris (L5, S1)	Laterally rotates thigh[c]; steadies femoral head in acetabulum

[a] The spinal cord segmental innervation is indicated (e.g., "S1, S2" means that the nerves supplying the piriformis are derived from the first two sacral segments of the spinal cord). Numbers in boldface (**S1**) indicate the main segmental innervation. Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

[b] The gemelli muscles blend with the tendon of the obturator internus as it attaches to the greater trochanter of the femur.

[c] There are six lateral rotators of the thigh: piriformis, obturator internus, superior and inferior gemelli, quadratus femoris, and obturator externus. These muscles also stabilize the hip joint.

Posterior Thigh Muscles

- Hamstring muscles ➢
  - Extend hip & flex knee •
  - Tibial division of sciatic nerve •
- Semitendinosus m. •
- Semimembranosus m. •
- Biceps femoris m. •
  - Long head •
  - Short head •
  - Fibular division of the sciatic nerve •

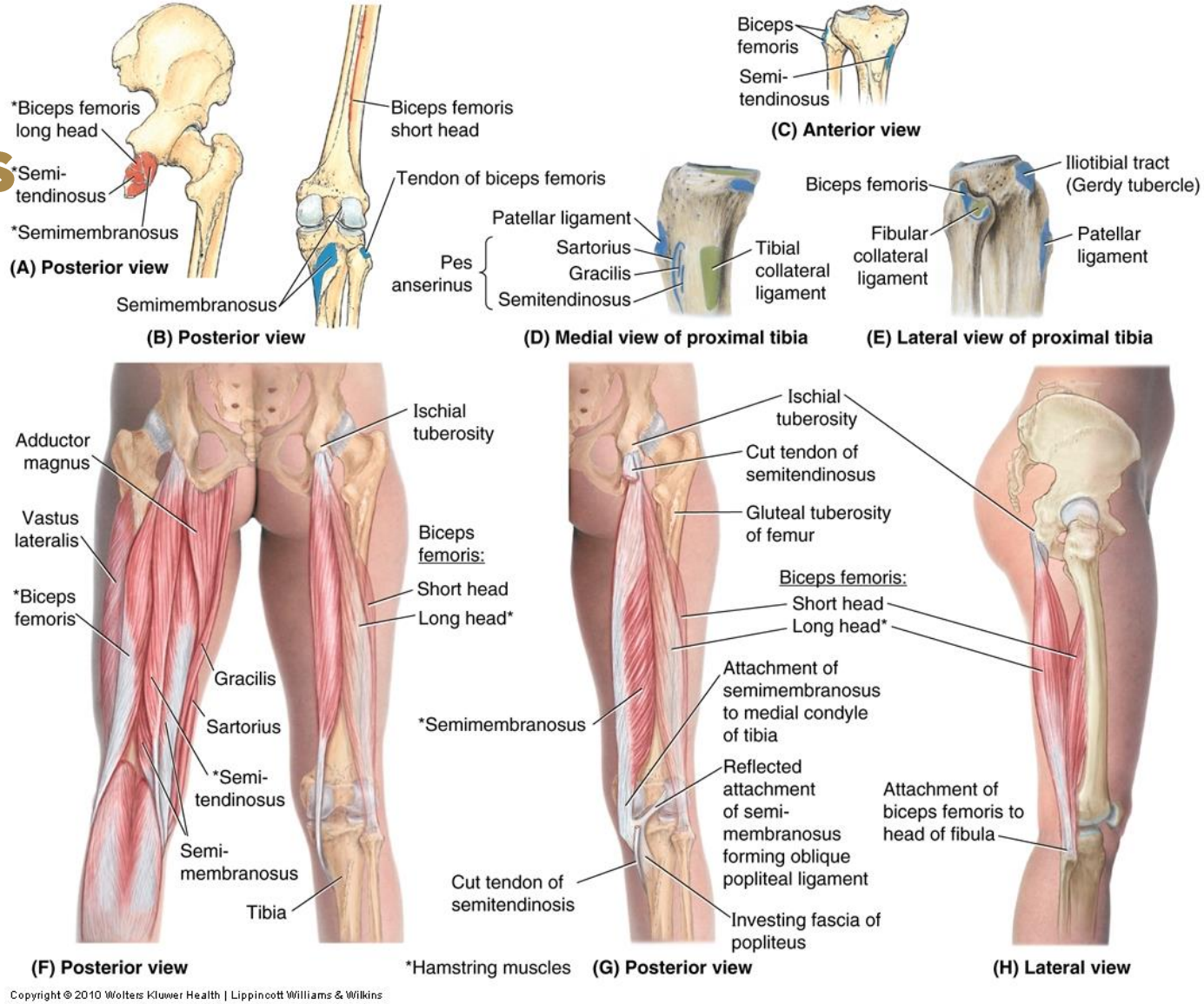


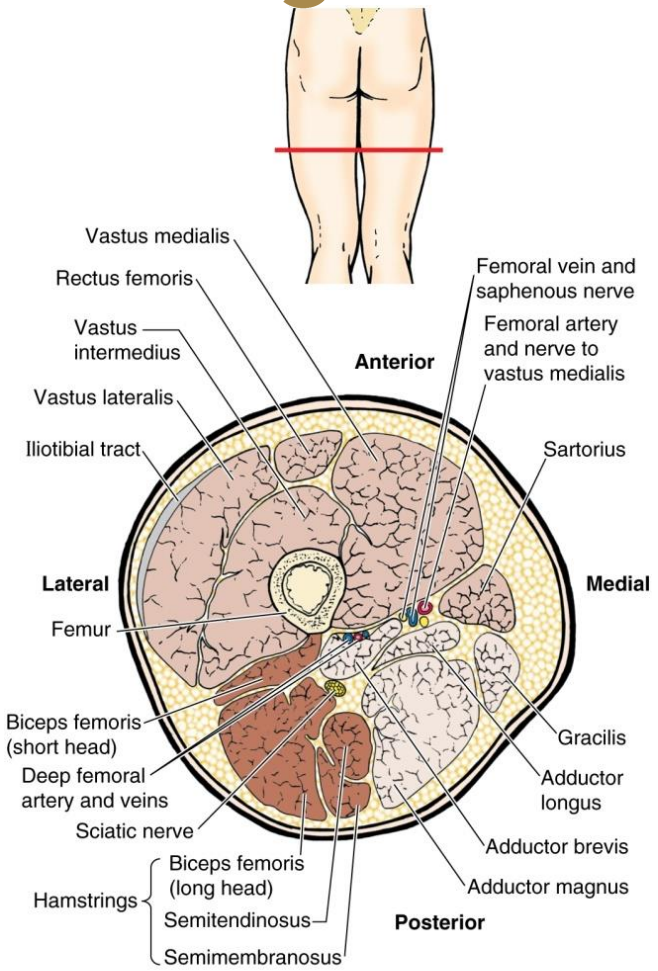
TABLE 5.7. MUSCLES OF POSTERIOR THIGH: EXTENSORS OF HIP AND FLEXORS OF KNEE

Muscle[a]	Proximal Attachment	Distal Attachment	Innervation[b]	Main Action
Semitendinosus	Ischial tuberosity	Medial surface of superior part of tibia	Tibial division of sciatic nerve part of tibia (**L5, S1,** S2)	Extend thigh; flex leg and rotate it medially when knee is flexed; when thigh and leg are flexed, these muscles can extend trunk
Semimembranosus		Posterior part of medial condyle of tibia; reflected attachment forms oblique popliteal ligament (to lateral femoral condyle)		
Biceps femoris	Long head: ischial tuberosity Short head: linea aspera and lateral supracondylar line of femur	Lateral side of head of fibula; tendon is split at this site by fibular collateral ligament of knee	Long head: tibial division of sciatic nerve (L5, **S1,** S2) Short head: common fibular division of sciatic nerve (L5, **S1,** S2)	Flexes leg and rotates it laterally when knee is flexed; extends thigh (e.g., accelerating mass during first step of gait).

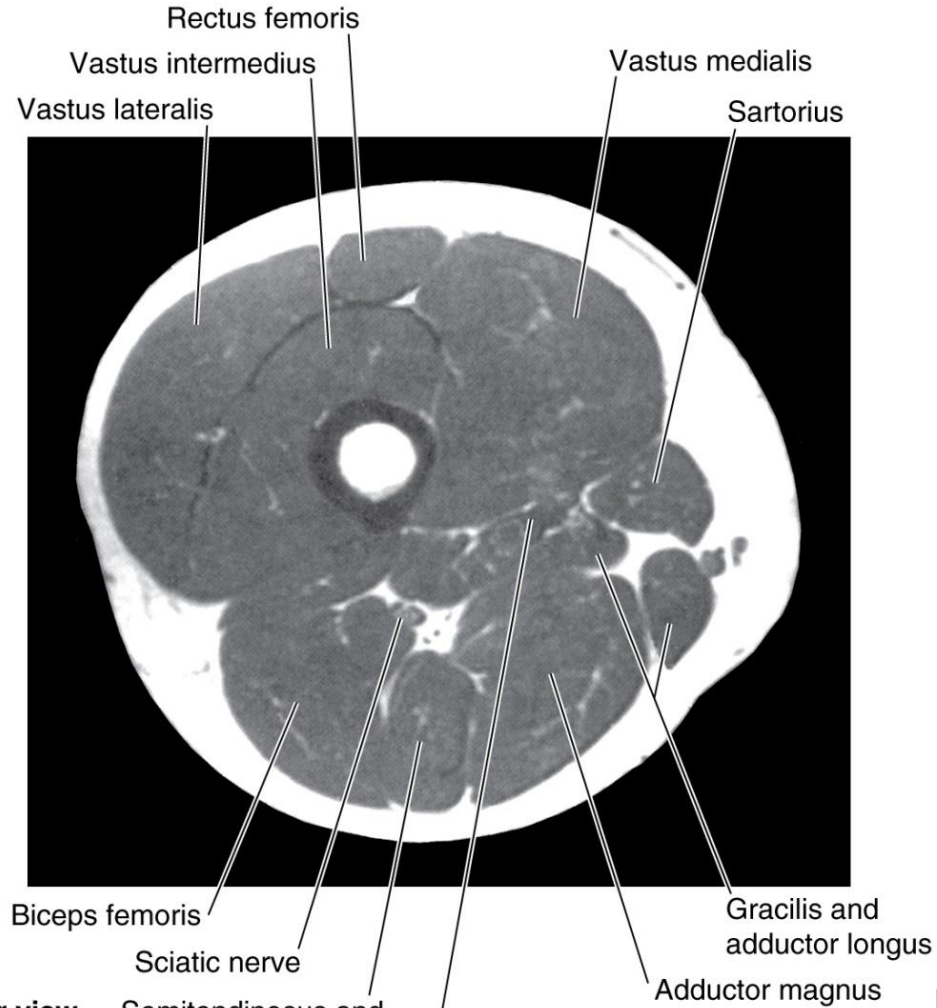
[a] Collectively these three muscles are known as hamstrings.

[b] The spinal cord segmental innervation is indicated (e.g., "L5, S1, S2" means that the nerves supplying the semitendinosus are derived from the fifth lumbar segment and first two sacral segments of the spinal cord). Numbers in boldface (**L5, S1**) indicate the main segmental innervation. Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

Thigh Muscles in Radiographic Image



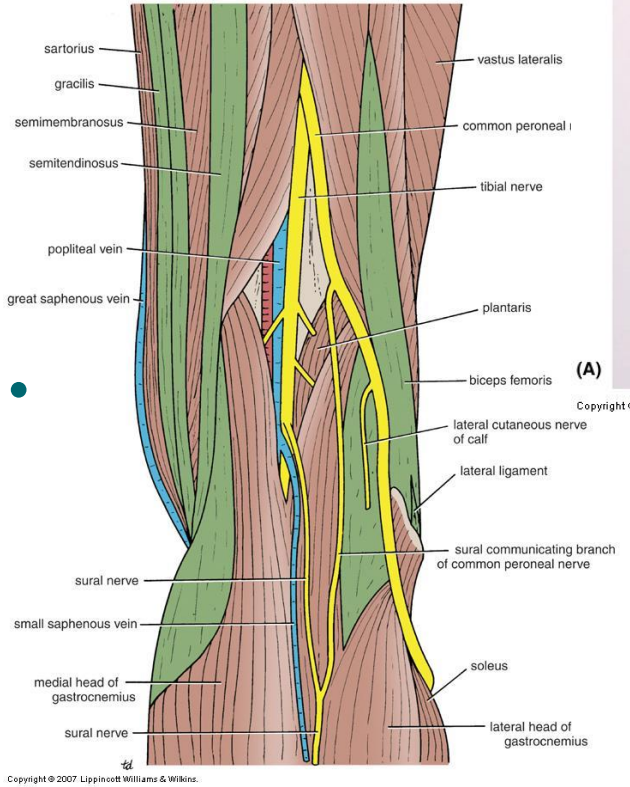
(B) Inferior view, right thigh



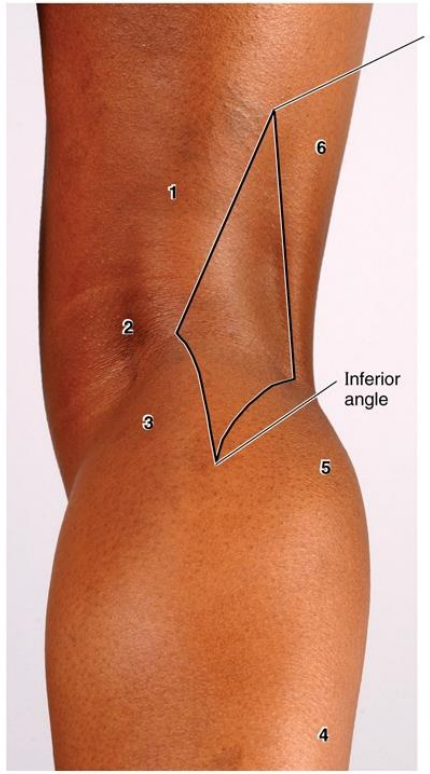
(C) Inferior view, right thigh

Popliteal fossa

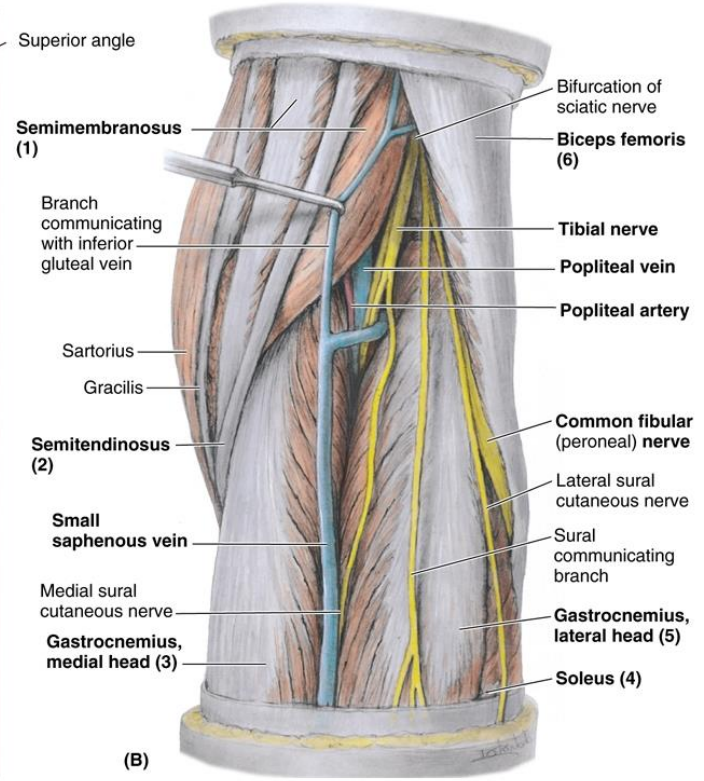
- Shape •
- Boundaries •
- Content •
  - Popliteal vessels •
  - Small saphenous vein •
  - Common peroneal & tibial nerves •
  - Lymph nodes •
  - Popliteal fascia •



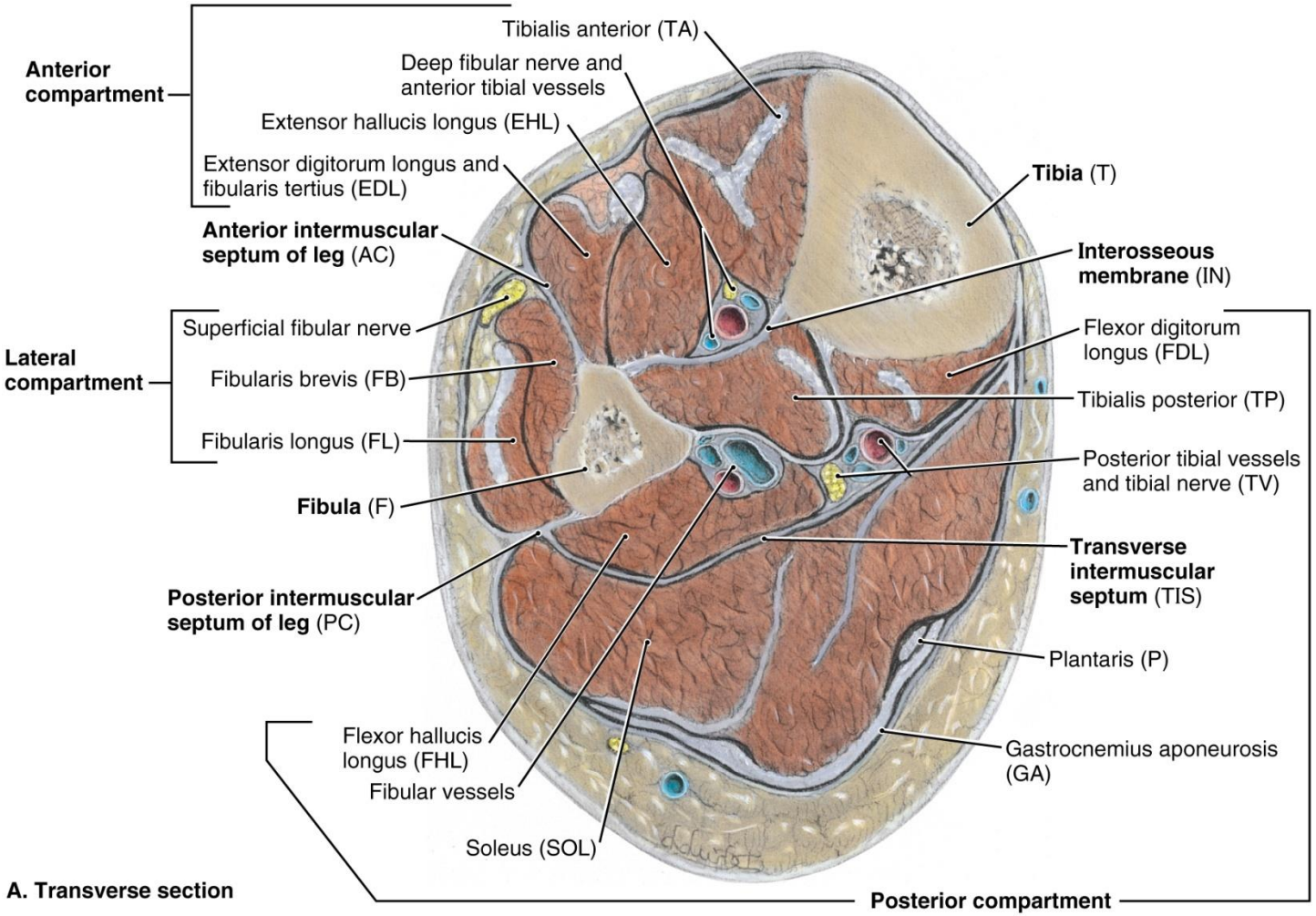
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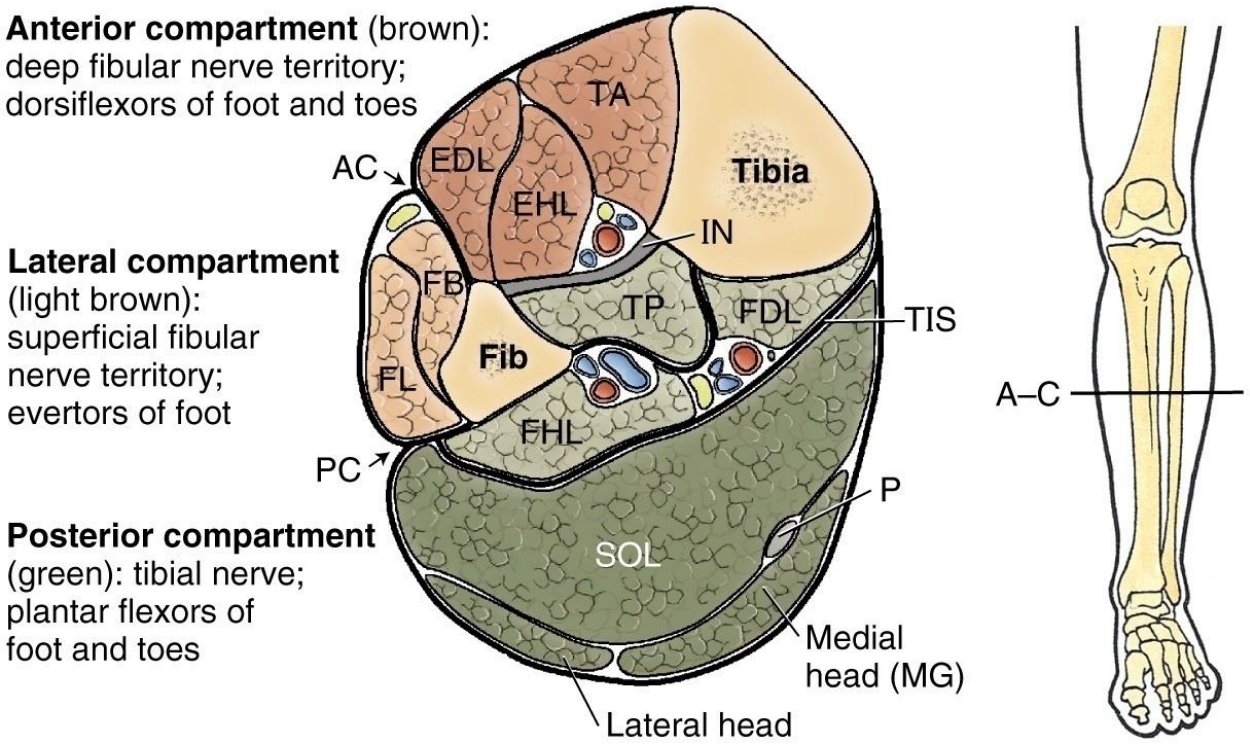


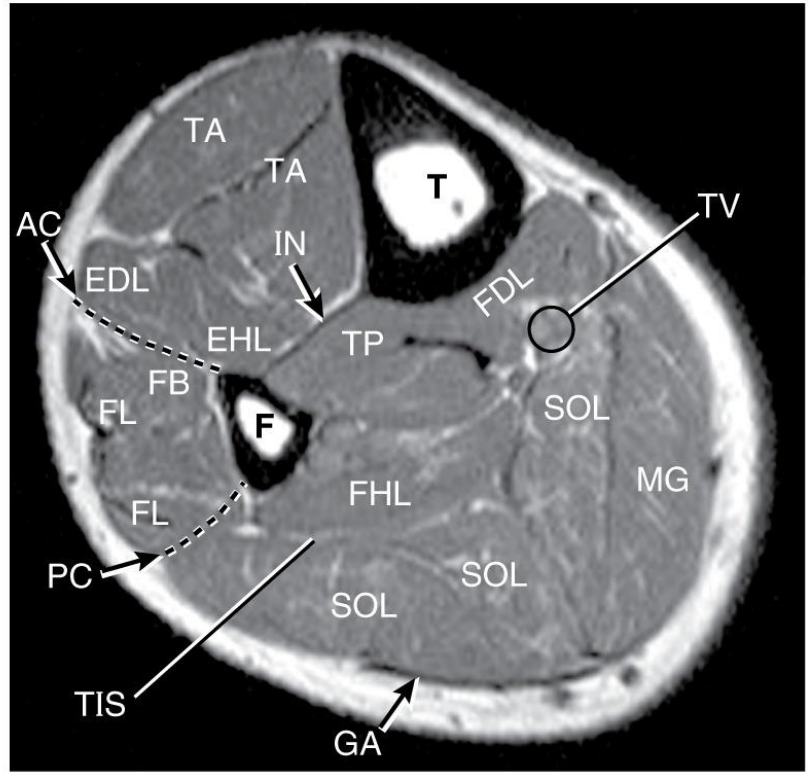
Leg compartments



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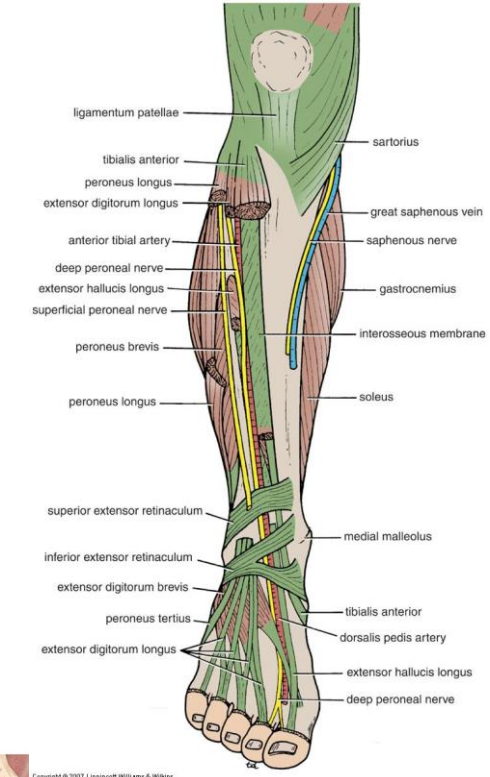
Leg compartments

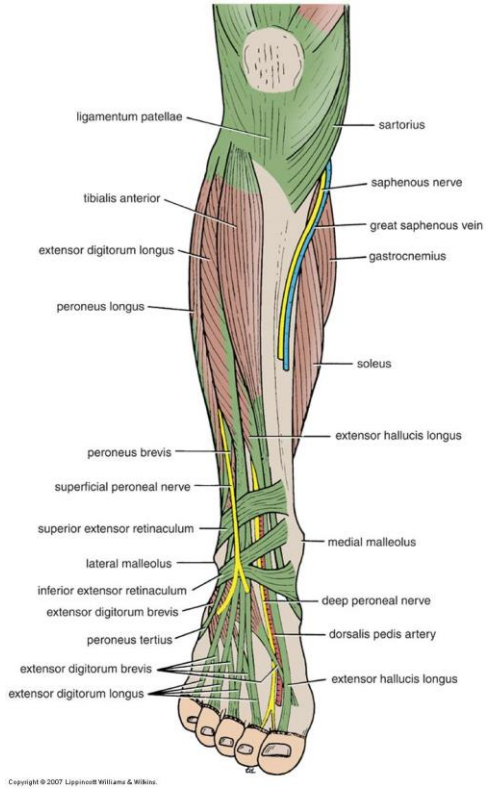


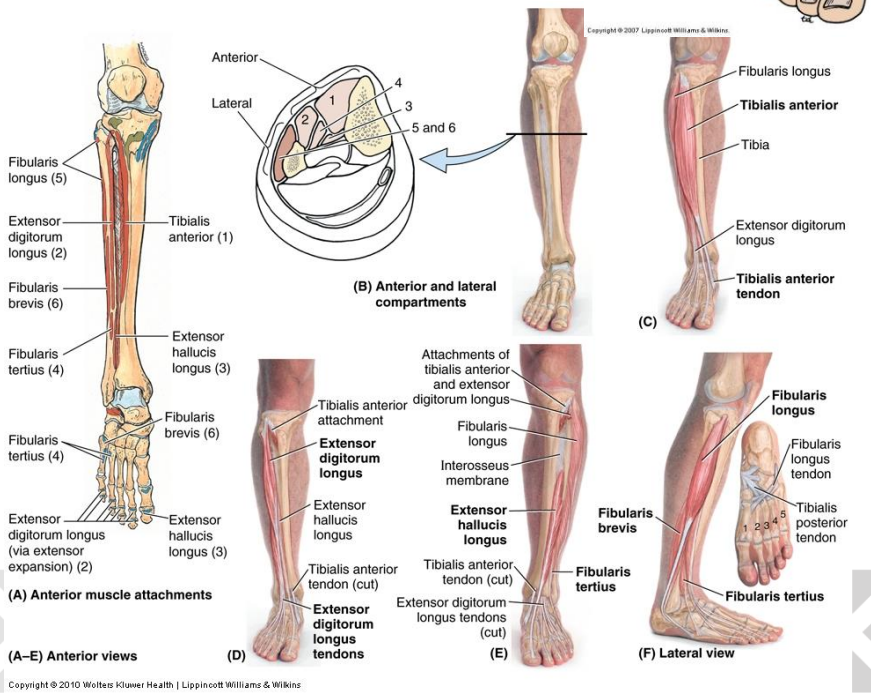


Muscles of Anterior Compartment of Leg

- Dorsiflexors of the ankle ➢
- Deep fibular nerve ➢
- Superior extensor retinaculum ❖
- Inferior extensor retinaculum ❖
- Tibialis anterior •
- Extensor digitorum longus •
- Extensor expansion (proximal phalanx)
  - Central band (middle phalanx) •
  - Lateral bands (distal phalanx) •
- Extensor hallucis longus •
- Fibularis tertius •







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Muscles of Lateral Compartment of Leg

Eversion of foot➢

Superficial fibular nerve➢

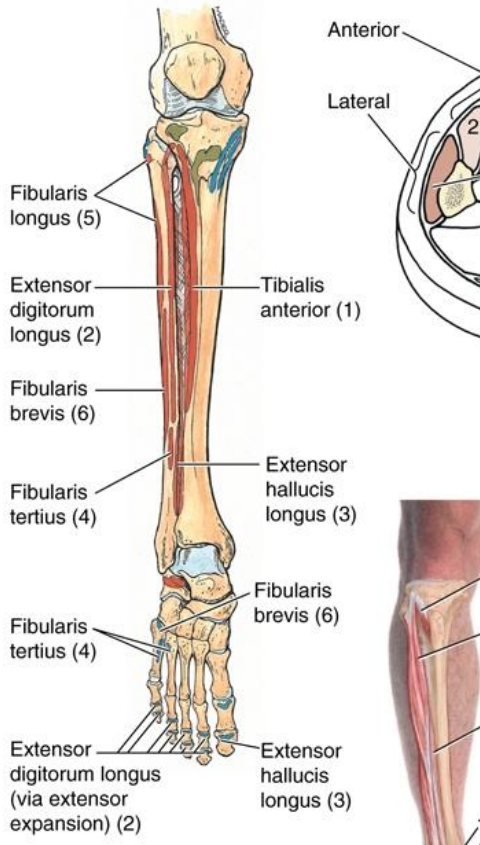
Superior peroneal retinaculum❖

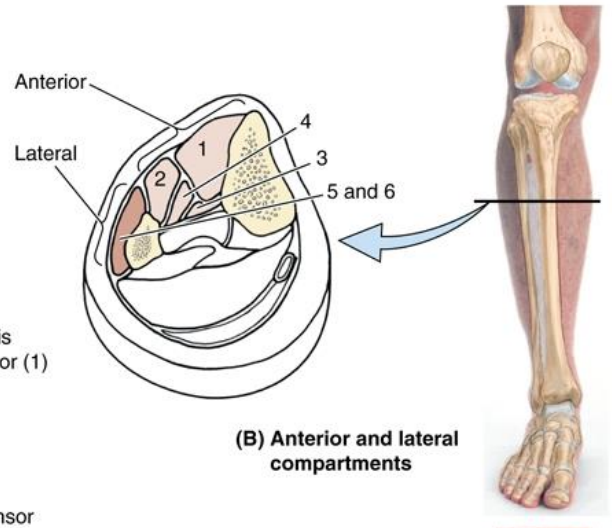
Inferior peroneal retinaculum❖

Peroneus longus m. •

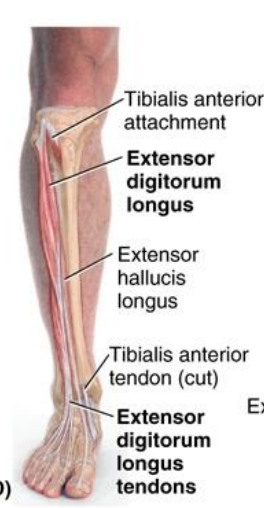
Longer & superficial •

Peroneus brevis m. •

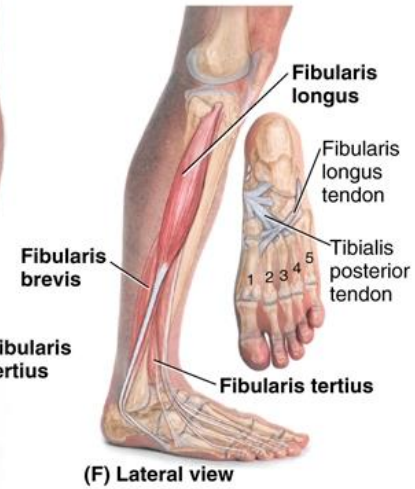




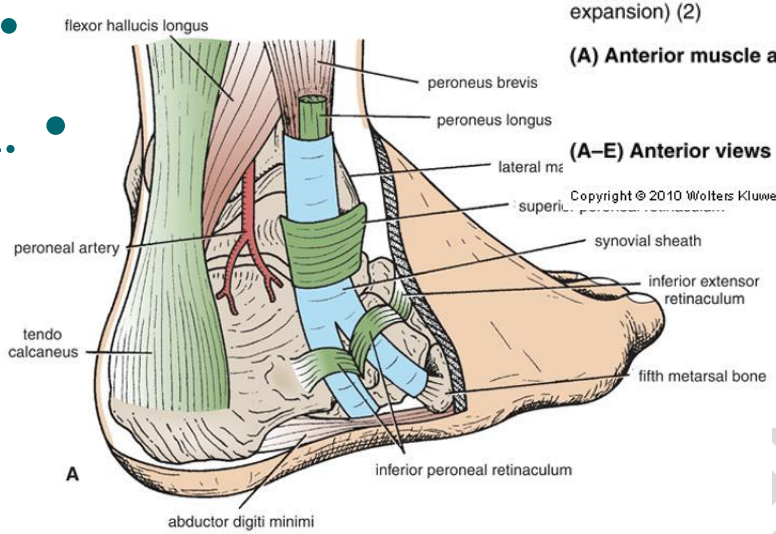








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TABLE 5.10. MUSCLES OF ANTERIOR AND LATERAL COMPARTMENTS OF LEG

Muscle[a]	Proximal Attachment	Distal Attachment	Innervation[b]	Main Action
Anterior compartment				
Tibialis anterior (1)	Lateral condyle and superior half of lateral surface of tibia and interosseous membrane	Medial and inferior surfaces of medial cuneiform and base of 1st metatarsal	Deep fibular nerve (**L4**, L5)	Dorsiflexes ankle and inverts foot
Extensor digitorum longus (2)	Lateral condyle of tibia and superior three quarters of medial surface of fibula and interosseous membrane	Middle and distal phalanges of lateral four digits		Extends lateral four digits and dorsiflexes ankle
Extensor hallucis longus (3)	Middle part of anterior surface of fibula and interosseous membrane	Dorsal aspect of base of distal phalanx of great toe (hallux)		Extends great toe and dorsiflexes ankle
Fibularis tertius (4)	Inferior third of anterior surface of fibula and interosseous membrane	Dorsum of base of 5th metatarsal		Dorsiflexes ankle and aids in eversion of foot
Lateral compartment				
Fibularis longus (5)	Head and superior two thirds of lateral surface of fibula	Base of 1st metatarsal and medial cuneiform	Superficial fibular nerve (**L5, S1,** S2)	Everts foot and weakly plantarflexes ankle
Fibularis brevis (6)	Inferior two thirds of lateral surface of fibula	Dorsal surface of tuberosity on lateral side of base of 5th metatarsal		

[a] Numbers refer to Figure 5.55A & B.

[b] The spinal cord segmental innervation is indicated (e.g., "L4, L5" means that the nerves supplying the tibialis anterior are derived from the fourth and fifth lumbar segments of the spinal cord). Numbers in boldface (**L4**) indicate the main segmental innervation. Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

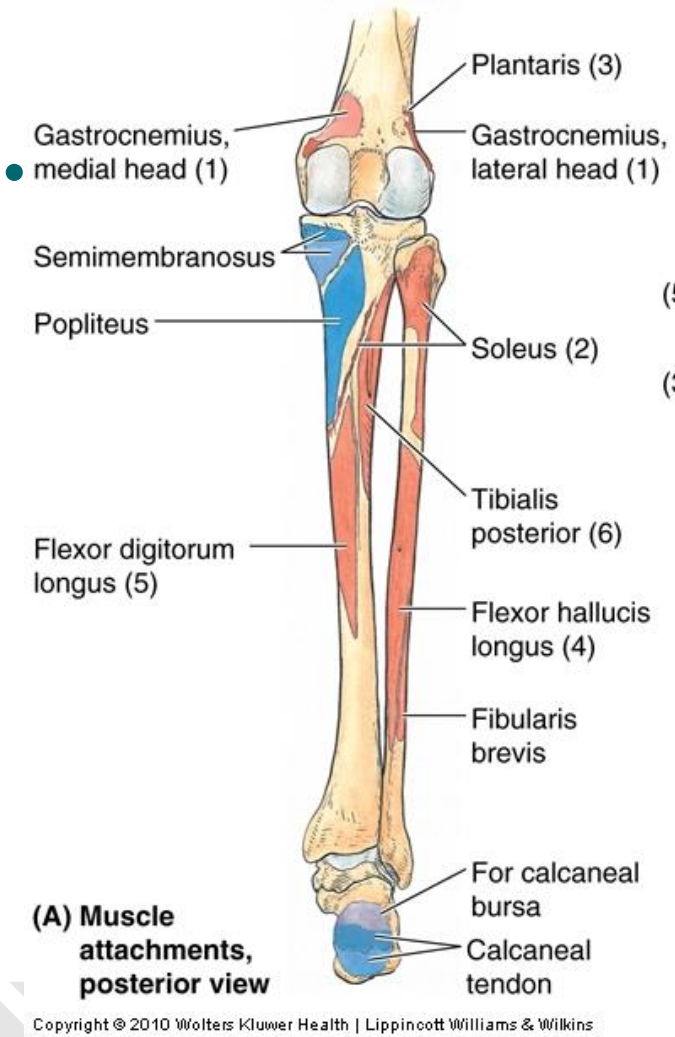
Muscles of the Posterior Compartment of Leg

Planter flexion➢

Tibial nerve➢

Superficial layer (calf muscles)

- Gastronemius •
  - Large, superficial •
  - Two heads •
  - Flex knee and ankle •
  - Calcaneal tendon •
- Soleus •
  - Calcaneal tendon •
- Plantaris •
  - Flex knee and ankle •
  - Traverse the knee articular capsule



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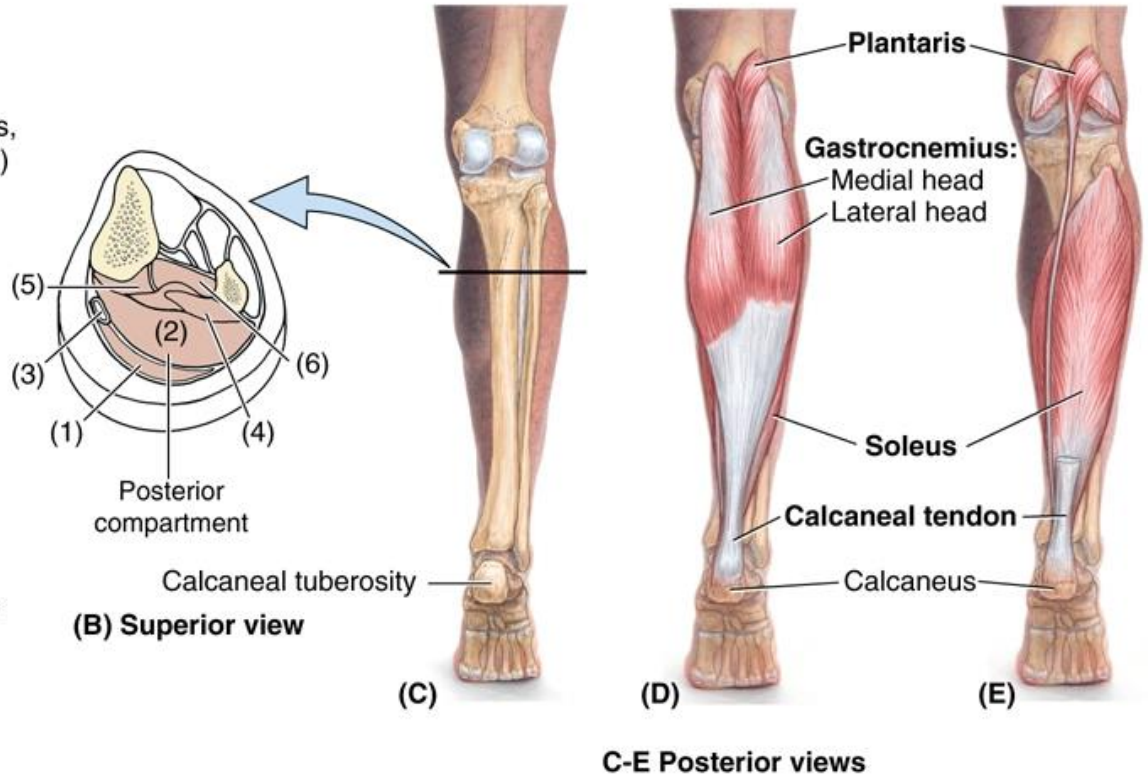


TABLE 5.13.I. SUPERFICIAL MUSCLES OF POSTERIOR COMPARTMENT OF LEG

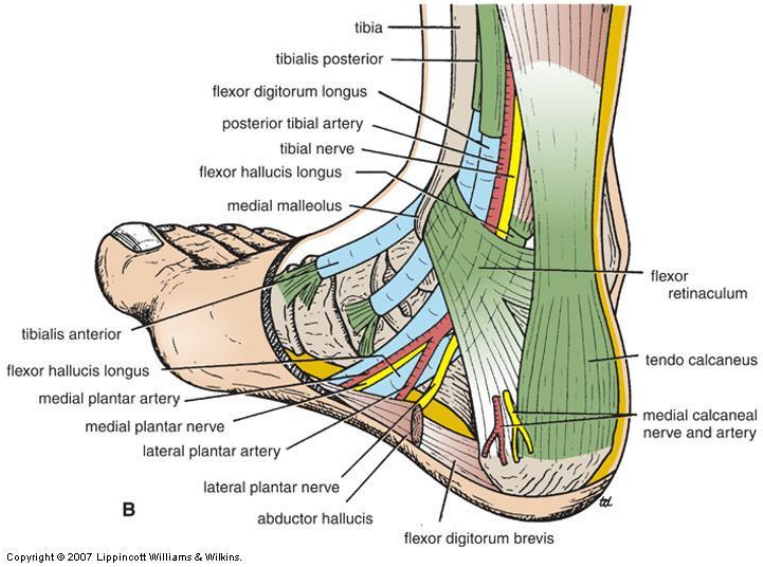
Muscle[a]	Proximal Attachment	Distal Attachment	Innervation[b]	Main Action
Gastrocnemius (1)	Lateral head: lateral aspect of lateral condyle of femur	Posterior surface of calcaneus via calcaneal tendon	Tibial nerve (S1, S2)	Plantarflexes ankle when knee is extended; raises heel during walking; flexes leg at knee joint
	Medial head: popliteal surface of femur; superior to medial condyle			
Soleus (2)	Posterior aspect of head and superior quarter of posterior surface of fibula; soleal line and middle third of medial border of tibia; and tendinous arch extending between the bony attachments			Plantarflexes ankle independent of position of knee; steadies leg on foot
Plantaris (3)	Inferior end of lateral supracondylar line of femur; oblique popliteal ligament			Weakly assists gastrocnemius in plantarflexing ankle

[a] Numbers refer to Figure 5.60A.

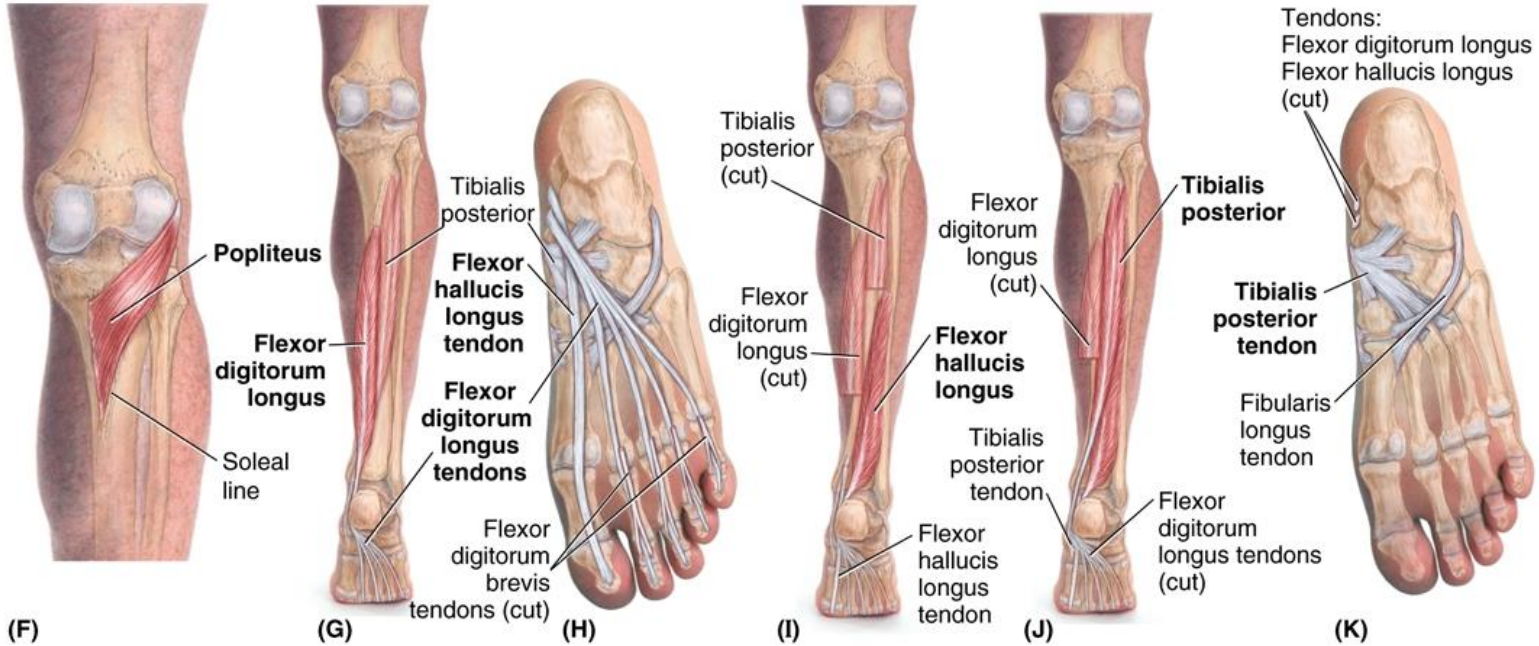
[b] The spinal cord segmental innervation is indicated (e.g., "S1, S2" means that the nerves supplying these muscles are derived from the first and second sacral segments of the spinal cord). Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

Muscles of the Posterior Compartment of Leg

- Deep layer •
  - Transverse intermuscular septum ➢
  - Flexor retinaculum ➢
  - Popliteus •
  - Flexor hallucis longus •
  - Flexor digitorum longus •
  - Tibialis posterior •
    - inversion •



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F, G, I, and J Posterior views

H and K Inferior views

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TABLE 5.13.II. DEEP MUSCLES OF POSTERIOR COMPARTMENT OF LEG

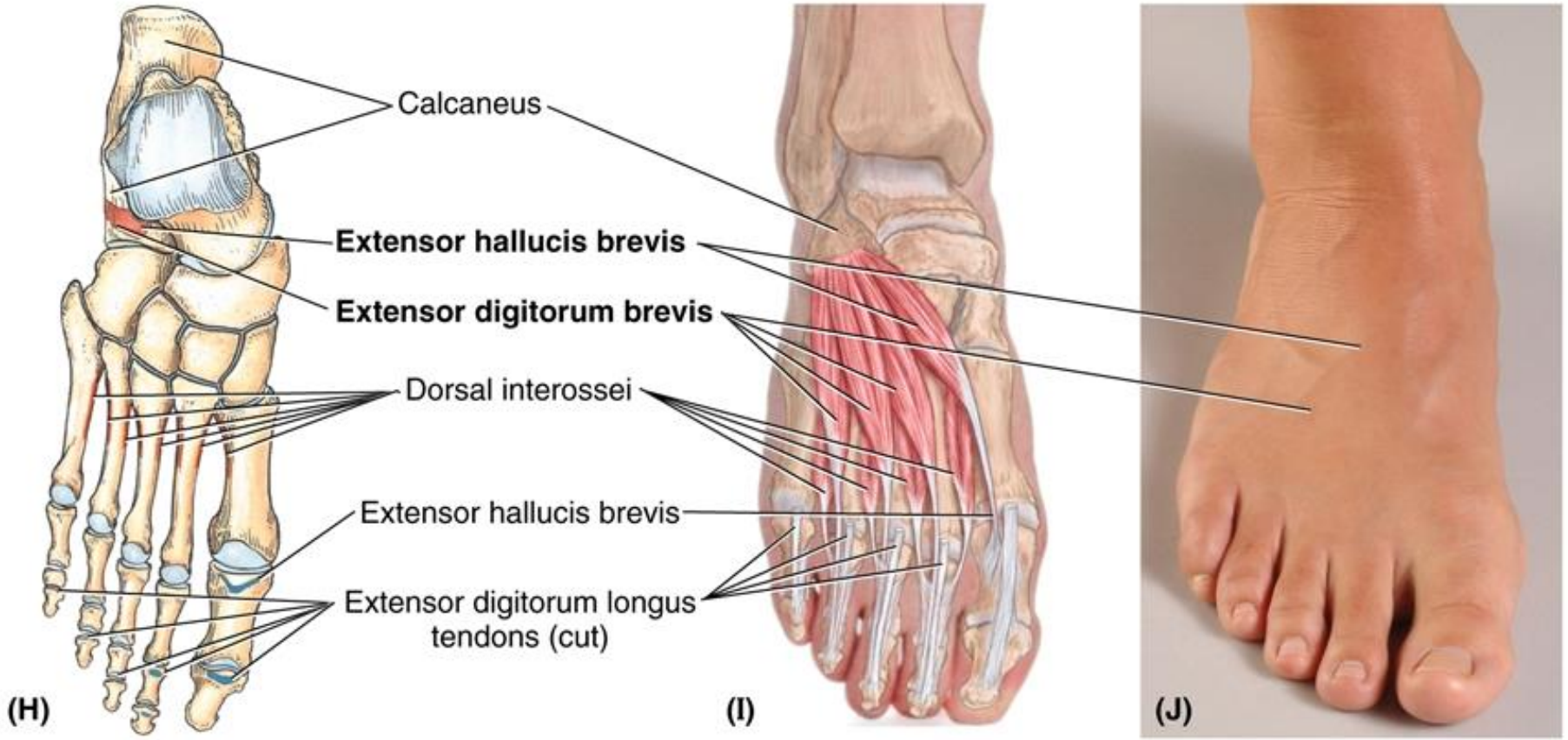
Muscle[a]	Proximal Attachment	Distal Attachment	Innervation[b]	Main Action
Popliteus	Lateral surface of lateral condyle of femur and lateral meniscus	Posterior surface of tibia, superior to soleal line	Tibial nerve (L4, L5, S1)	Weakly flexes knee and unlocks it by rotating femur 5° on fixed tibia; medially rotates tibia of unplanted limb
Flexor hallucis longus (4)	Inferior two thirds of posterior surface of fibula; inferior part of interosseous membrane	Base of distal phalanx of great toe (hallux)	Tibial nerve (S2, S3)	Flexes great toe at all joints; weakly plantarflexes ankle; supports medial longitudinal arch of foot
Flexor digitorum longus (5)	Medial part of posterior surface of tibia inferior to soleal line; by a broad tendon to fibula	Bases of distal phalanges of lateral four digits		Flexes lateral four digits; plantarflexes ankle; supports longitudinal arches of foot
Tibialis posterior (6)	Interosseous membrane; posterior surface of tibia inferior to soleal line; posterior surface of fibula	Tuberosity of navicular, cuneiform, cuboid, and sustentaculum tali of calcaneus; bases of 2nd, 3rd, and 4th metatarsals	Tibial nerve (L4, L5)	Plantarflexes ankle; inverts foot

[a] Numbers refer to Figure 5.60A.

[b] The spinal cord segmental innervation is indicated (e.g., "S2, S3" means that the nerves supplying the flexor hallucis longus are derived from the second and third sacral segments of the spinal cord). Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

Muscles of the Dorsum of the Foot

- Extensor digitorum brevis m.
- Extensor hallucis brevis m.



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TABLE 5.14.III. MUSCLES OF FOOT: DORSUM OF FOOT

Muscle	Proximal Attachment	Distal Attachment	Innervation[a]	Main Action
Extensor digitorum brevis	Calcaneus (floor of tarsal sinus); interosseous talocalcaneal ligament; stem of inferior extensor retinaculum	Long extensor tendons of four medial digits (toes 2–4)	Deep fibular nerve (L5 or S1, or both)	Aids the extensor digitorum longus in extending the four medial toes at the metatarsophalangeal and interphalangeal joints
Extensor hallucis brevis	In common with extensor digitorum brevis (above)	Dorsal aspect of base of proximal phalanx of great toe (digit 1)		Aids the extensor hallucis longus in extending the great toe at the metatarsophalangeal joint

[a]The spinal cord segmental innervation is indicated (e.g., "L5 or S1" means that the nerve supplying the extensor digitorum brevis is derived from either the fifth lumbar segment or first sacral segment of the spinal cord). Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

Muscles of the Sole of the Foot

1st layer •

Abductor hallucis m. •

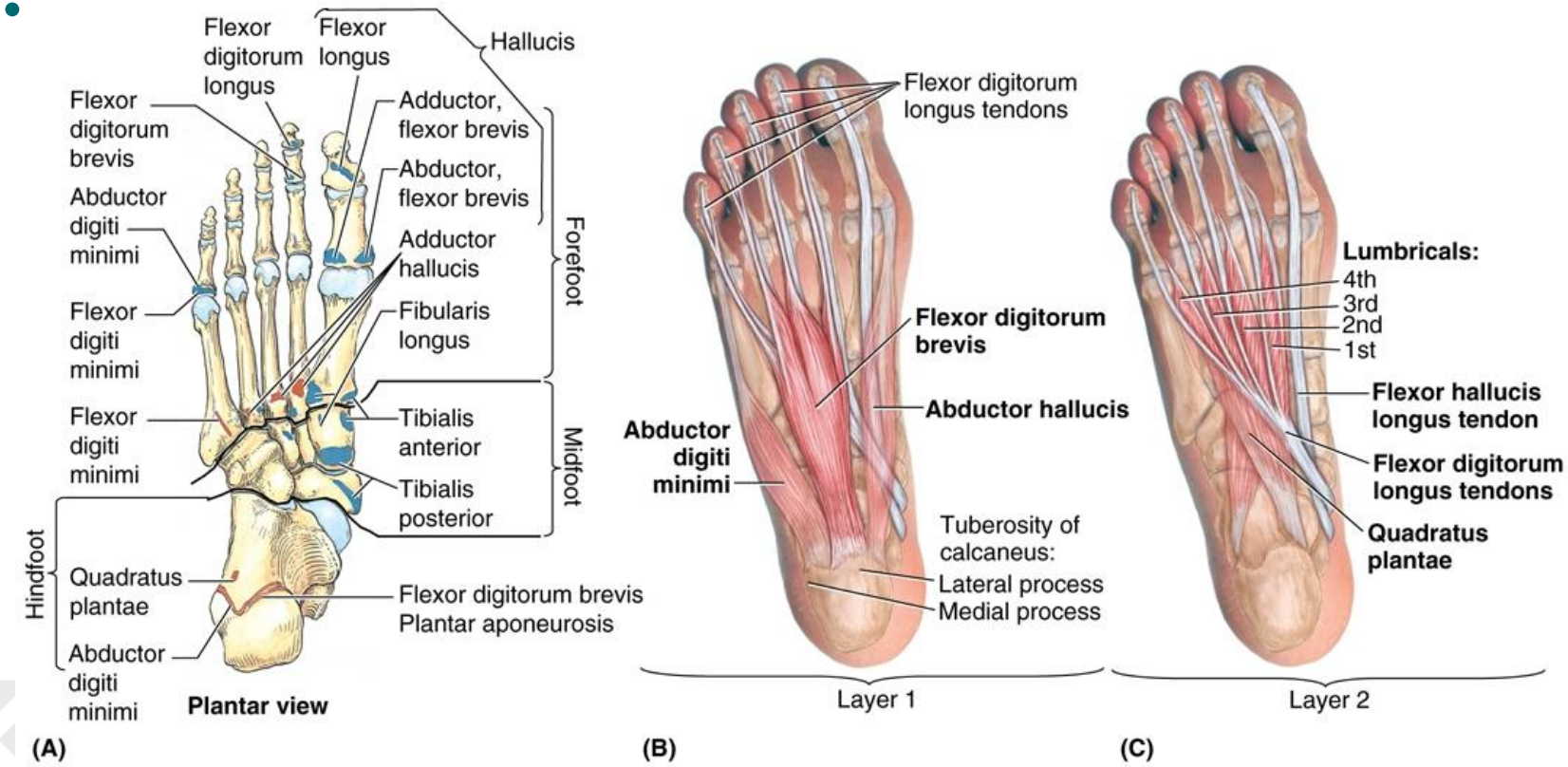
Flexor digitorum brevis m. – middle phalanges •

Abductor digiti minimi m. •

2nd layer •

Quadratus plantae m. – fixes the flexor digitorum longus tendons •

Lumbricals •

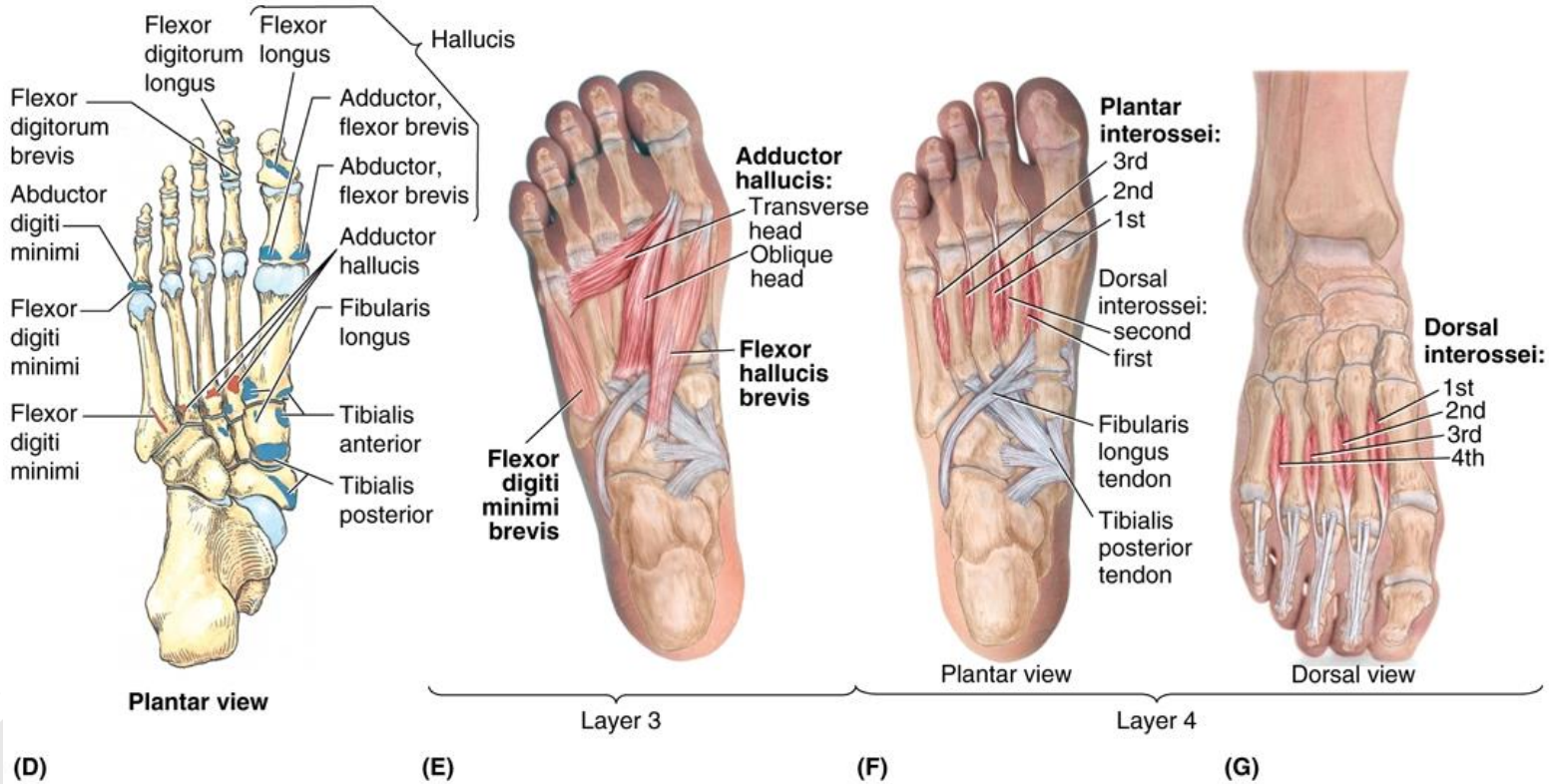


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Muscles of the Sole of the Foot

3rd layer •
Flexor hallucis brevis m. – two sesamoid bones •
Adductor hallucis m. – transverse & oblique heads •
Flexor digiti minimi brevis m. •

4th layer •
Planter interossei mm. – 1st-3rd (adductors) •
Dorsal interossei mm. – 1st- 4th (abductors) •



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TABLE 5.14.I. MUSCLES OF FOOT: 1ST AND 2ND LAYERS OF SOLE

Muscle	Proximal Attachment	Distal Attachment	Innervation[a]	Main Action[b]
1st layer				
Abductor hallucis	Medial tubercle of tuberosity of calcaneus; flexor retinaculum; plantar aponeurosis	Medial side of base of proximal phalanx of 1st digit	Medial plantar nerve (S2, S3)	Abducts and flexes 1st digit (great toe, hallux)
Flexor digitorum brevis	Medial tubercle of tuberosity of calcaneus; plantar aponeurosis; intermuscular septa	Both sides of middle phalanges of lateral four digits	Medial plantar nerve (S2, S3)	Flexes lateral four digits
Abductor digiti minimi	Medial and lateral tubercles of tuberosity of calcaneus; plantar aponeurosis; intermuscular septa	Lateral side of base of proximal phalanx of 5th digit	Lateral plantar nerve (S2, S3)	Abducts and flexes little toe (5th digit)
2nd layer				
Quadratus plantae	Medial surface and lateral margin of plantar surface of calcaneus	Posterolateral margin of tendon of flexor digitorum longus	Lateral plantar nerve (S2, S3)	Assists flexor digitorum longus in flexing lateral four digits
Lumbricals	Tendons of flexor digitorum longus	Medial aspect of expansion over lateral four digits	Medial one: medial plantar nerve (S2, S3) Lateral three: lateral plantar nerve (S2, S3)	Flex proximal phalanges, extend middle and distal phalanges of lateral four digits

[a] The spinal cord segmental innervation is indicated (e.g., "S2, S3" means that the nerves supplying the abductor hallucis are derived from the second and third sacral segments of the spinal cord). Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

[b] Despite individual actions, the primary function of the intrinsic muscles of the sole of the foot is to resist flattening or maintain the arch of the foot.

TABLE 5.14.II. MUSCLES OF FOOT: 3RD AND 4TH LAYERS OF SOLE

Muscle	Proximal Attachment	Distal Attachment	Innervation[a]	Main Action[b]
3rd layer				
Flexor hallucis brevis	Plantar surfaces of cuboid and lateral cuneiforms	Both sides of base of proximal phalanx of 1st digit	Medial plantar nerve (S2, S3)	Flexes proximal phalanx of 1st digit
Adductor hallucis	Oblique head: bases of metatarsals 2–4	Tendons of both heads attach to lateral side of base of proximal phalanx of 1st digit	Deep branch of lateral plantar nerve (S2, S3)	Traditionally said to adduct 1st digit; assists in transverse arch of foot by metatarsals medially
	Transverse head: plantar ligaments of metatarsophalangeal (MTP) joints			
Flexor digit minimi brevis	Base of 5th metatarsal	Base of proximal phalanx of 5th digit	Superficial branch of lateral plantar nerve (S2, S3)	Flexes proximal phalanx of 5th digit, thereby assisting with its flexion
4th layer				
Plantar interossei (three muscles)	Bases and medial sides of metatarsals 3–5	Medial sides of bases of phalanges of 3rd–5th digits	Lateral plantar nerve (S2, S3)	Adduct digits (2–4) and flex metatarsophalangeal joints
Dorsal interossei (four muscles)	Adjacent sides of metatarsals 1–5	1st: medial side of proximal phalanx of 2nd digit;	Lateral plantar nerve (S2, S3)	Abduct digits (2–4) and flex metatarsophalangeal joints
		2nd–4th: lateral sides of 2nd–4th digits		

[a] The spinal cord segmental innervation is indicated (e.g., "S2, S3" means that the nerves supplying the flexor hallucis brevis are derived from the second and third sacral segments of the spinal cord). Damage to one or more of the listed spinal cord segments or to the motor nerve roots arising from them results in paralysis of the muscles concerned.

[b] Despite individual actions, the primary function of the intrinsic muscles of the sole of the foot is to resist flattening or maintain the arch of the foot.

جامعة أم القرى
UMM AL-QURA UNIVERSITY

Thanks